Afloat or Adrift: A Longitudinal Study of Latent Personality Profiles and Future-of-Work Skills Across Midlife

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This article reports a secondary analysis of the Midlife in the United States (MIDUS) core panel (M1 1995-96, M2 2004-06, M3 2013-14) and the independent MIDUS Refresher sample. MIDUS data-use disclosure appears in Appendix B. The design and analyses were not preregistered. As a secondary analysis of de-identified, publicly available data, the study involved no new data collection from human participants and did not constitute human-subjects research requiring institutional review board approval. This research received no external funding, and the author has no conflicts of interest to declare. A preprint of this article is available on the PsyArXiv preprint server. Generative artificial intelligence tools were used to assist with drafting, editing, and analysis-code review; the author verified and takes responsibility for all content.

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Abstract

A pressing question during the current artificial intelligence (AI)-driven economic transformation is whether workers will develop and retain AI-related workforce skills. Those skills rest on psychological resources, yet little longitudinal evidence tracks whether workers recover or disengage when those resources are strained. In a secondary analysis of the Midlife in the United States (MIDUS) national panel (N = 7,108; approximately 20 years) and independent Refresher (N = 3,577), latent personality profiles were enumerated from Big Five and Agency indicators. A joint latent Markov model of profile transitions was estimated. Study 1 addressed research questions on latent profile structure, replication, and incremental validity. Study 2 tested six hypotheses on transition structure, transition pathways, skill-criterion change, and self-determination theory predictors of transition direction. Four profiles were supported: Resilient (36.4%), Distressed (29.5%), Reserved (29.0%), and Antagonistic (5.1%). Resilient members reported highest psychosocial skills; Antagonistic members reported highest income, prestige, and scored highest on analytic performance. Profile membership showed no incremental predictive validity beyond continuous traits. Distressed profile membership declined from 29.5% to 7.2% across two decades via recovery (movement to Resilient) and disengagement (movement to Reserved) pathways. Leaving the Distressed profile was associated with lower health-related lost productive time (recovered movers -$3,049 per worker-year, 95% CI [-$5,038, -$1,060]). Purpose predicted recovery- versus disengagement-oriented transition (OR = 1.23 per SD, 95% CI [1.01, 1.50]). Event-sampled diary data (N = 2,314) corroborated profile interpretation. Implications for sustaining workforce skills during AI-era economic transition are discussed.

Keywords: latent profile analysis, latent transition analysis, self-determination theory, purpose in life, future of work, MIDUS

The evidence that work is entering a structural transition rather than a cyclical adjustment is by now substantial. Task-level analyses estimate that a large share of occupations will have their core activities reorganized by Artificial Intelligence (AI) (Acemoglu & Restrepo, 2019; Brynjolfsson & McAfee, 2014; Frey & Osborne, 2017) and employer surveys concur. In the World Economic Forum's most recent Future of Jobs report, roughly two in five of workers' existing skill sets are projected to be transformed or outdated within five years, and AI and information-processing technology are the development employers most widely expect to transform their business (World Economic Forum, 2025). Economic historians document that technological transitions redistribute psychological strain as well as tasks and wages, concentrating distress in the populations whose occupational roles are most disrupted (Autor, 2015; Case & Deaton, 2015). In the World Economic Forum's (2025) Future of Jobs Report, employers rank analytical thinking, resilience and agility, leadership and social influence, creative thinking, curiosity and lifelong learning, and motivation and self-awareness among the core skills judged essential today and projected to rise through 2030 as employers adapt workforces to AI-driven change. The question posed by this transition is therefore not only which skills employers will prioritize, but what determines whether workers under transition-induced strain emerge with those skills intact.

Workers whose roles are destabilized may recover control and re-engage adaptive and prosocial functioning, or they may withdraw effort, development, and interpersonal investment while remaining employed. A robust and long-documented response to sustained work strain is disengagement—the withdrawal of effort, energy, and interpersonal investment that follows when chronic demands deplete personal and job resources (Demerouti et al., 2001; Lee & Ashforth, 1996; Maslach & Leiter, 2016; Sverke et al., 2002). If economy-wide transition

increases occupational strain then retaining workforce skills may depend less on the prevalence of distress than on the distribution of recovery versus disengagement during distress. Existing evidence on a recovery versus disengagement pathway during distress is largely variable-centered, short-term, and outcome-specific (Gallup, 2024; Shoss, 2017). Less is known about how multivariate personality profiles transition over the multiyear intervals that structural labor-market change requires (cf. Howard et al., 2020; Kleshinski et al., 2024).

The aforementioned workforce skills are built on personality traits and psychological resources. Whether workers keep, build, or lose them under strain is what determines whether the skills endure. Assessing how traits and resources change over time requires long-term longitudinal data with well-validated psychological indicators (Roberts & Mroczek, 2008; Specht et al., 2014). The current study spans an interval long enough to observe that change and to watch distressed configurations resolve.

AI-driven automation is likely to affect the psychological resources necessary to acquire and maintain skills. Self-determination theory (SDT) traces these resources to the satisfaction of three basic psychological needs (BPNs): autonomy, competence, and relatedness (Deci & Ryan, 2000). For example, algorithmic management and automation that remove decision latitude reduce workers' volitional control over how work is done, undermining autonomy (Gagné et al., 2022; Kellogg et al., 2020; Parker & Grote, 2022). Opaque automation and displacement risk erode workers' beliefs that outcomes depend on their own action, reducing sense of control and mastery (Parker & Grote, 2022). Perceived control is among the most robust predictors of health and functioning across the lifespan (Lachman & Weaver, 1998). Technology-mediated coordination, algorithmic task allocation, and remote work arrangements can reduce the interpersonal contact and collegial support from which workplace relationships are built,

attenuating feelings of relatedness (Gagné et al., 2022; Golden et al., 2008; Kellogg et al., 2020; Parker & Grote, 2022). Finally, automation that strips decision latitude and task significance can inadvertently dismantle the job-design features from which perceived meaning is built (Hackman & Oldham, 1976; Steger et al., 2012). Because purpose and meaning are conceptually adjacent to the three needs (Martela et al., 2018), the present study examines purpose in life alongside the basic needs as a psychological resource that need-thwarting job design may likewise erode. Scholars have argued explicitly that the future of work will be decided partly by whether technology-mediated job design supports or frustrates basic needs (Gagné et al., 2022).

Latent Personality Profiles and Future-of-Work Skills

Most evidence linking personality to work outcomes uses a variable-centered approach in which traits are analyzed as independent predictors (Barrick & Mount, 1991; Judge et al., 2002; Ones et al., 2007). Person-centered mixture models instead test whether multivariate trait scores cluster into latent profiles with distinct configural patterns (Bergman & Magnusson, 1997; Meyer & Morin, 2016; Spurk et al., 2020; Yin et al., 2021). In latent profile analyses of the five-factor model (FFM) studies most often recover three- or four-class solutions anchored to Resilient, Overcontrolled, and Undercontrolled (ROU) prototypes (Asendorpf et al., 2001; Block & Block, 1980; Donnellan & Robins, 2010; Robins et al., 1996). Larger samples and alternative mixture specifications frequently add withdrawn or low-agreeableness classes beyond the ROU triad, for example, self-centered configurations and large average or reserved majorities (Favini et al., 2018; Gerlach et al., 2018; Isler et al., 2017). Optimal class count, however, varies with sample size, culture, and fit criteria (Specht et al., 2014; Yin et al., 2021). Prior profile work has also extended the personality indicator set beyond the five factors, for example by modeling the six HEXACO dimensions in latent profile analysis (Daljeet et al., 2017; Ferguson & Hull, 2017;

Gerlach et al., 2018). Latent profiles built on an explicit agentic dimension, as used here, remain comparatively rare, though the agentic dimension has been modeled alongside the Big Five in MIDUS and linked to consequential midlife health and well-being outcomes (Human et al., 2013). The present profiles are estimated from the five FFM traits plus Agency, a dimension of instrumental self-direction and proactive control that is especially salient when automation and algorithmic management constrain workers' autonomy over how work is done (Bakan, 1966; Parker & Grote, 2022); these six-dimension profiles are referred to throughout as latent personality profiles rather than FFM profiles. Three questions are investigated in the current study: (a) whether latent personality profiles replicate in a longitudinal panel measured with a common Big Five and Agency battery across two decades; (b) whether profile membership is associated with core- and rising-skills criteria defined by the World Economic Forum (2025); and (c) whether profiles add predictive value beyond continuous traits. In accord with convention (e.g., Costa et al., 2002; Spurk et al., 2020) profile description and criteria prediction are examined independently so that incremental validity is assessed independently.

Overview of the Present Research

The present research was designed to contribute to person-centered personality and SDT research in three related ways. First, the present study adopts a person-centered mixture-model approach to identify latent personality profiles from multivariate Big Five and Agency indicators, following recommendations for model enumeration, profile characterization, and construct validation (Meyer & Morin, 2016; Spurk et al., 2020; Yin et al., 2021). Second, the study evaluates whether profile membership adds explanatory value beyond continuous traits using incremental validity tests, in line with prior critiques of personality typologies (Costa et al., 2002; Donnellan & Robins, 2010). Third, the study extends cross-sectional latent profile work

with a joint latent Markov model to estimate profile transitions across three waves. These profile transitions assess whether the basic psychological needs predict transition direction among Distressed movers (Collins & Lanza, 2010; Howard et al., 2016, 2020; Kleshinski et al., 2024; Slaughter et al., 2021; Vaziri et al., 2020). Study 2 hypotheses were developed after the four-class solution and profile labels were established in Study 1, following the two-study sequence used when hypothesis tests depend on an initial person-centered solution (Gabriel et al., 2015; Howard et al., 2020; Kanitz et al., 2024; Zhang et al., 2024). This substantive-methodological design parallels recent works that discover and replicate profiles before testing antecedents, outcomes, or transitions (Kanitz et al., 2024; Kleshinski et al., 2024; Zhang et al., 2024) as well as recent organizational research on motivational configurations (Howard et al., 2016, 2020).

The outcome of the incremental-validity test determined whether profile findings would be interpreted as evidence of predictive superiority or as descriptive developmental structure. Study 2 estimated latent profile and latent transition structure, compared recovery and disengagement transitions out of the Distressed profile, tested SDT need predictors of transition direction, and integrated event-sampled daily diary data as a corroborative quantitative lens.

The current study contributes to longitudinal person-centered research (Gabriel et al., 2015; Kleshinski et al., 2024) by adding an independent replication sample, classification agreement across independent maximum-likelihood implementations, and a comparatively long (i.e., twenty-year) longitudinal interval. These features support external replication (Meyer & Morin, 2016; Spurk et al., 2020).

Self-Determination Theory: Basic Psychological Needs

Self-determination theory holds that autonomy, competence, and relatedness are BPNs whose satisfaction increases individual personality integration and well-being (Deci & Ryan,

2000; Ryan & Deci, 2017). The MIDUS datasets contain validated measures of each basic need. The Ryff Psychological Well-Being (PWB) autonomy and positive relations with others subscales were used as assessments of autonomy and relatedness (Ryff, 1989; Ryff & Keyes, 1995). Personal mastery was used to assess competence (Lachman & Weaver, 1998). Purpose in life, a eudaimonic well-being resource conceptually adjacent to the basic needs rather than a need itself, was also measured with the Ryff purpose subscale and treated as a predictor of transition direction (Martela et al., 2018; Ryff, 1989). Finally, workplace decision authority provides a contextual indicator of autonomy support in the job-design tradition (Karasek, 1979).

The degree to which the three BPNs were met was theorized to have two effects during profile transitions. The first is the likelihood of transition, or whether a person leaves the Distressed profile. The second is the direction of transition, or which pathway movers take when they leave the Distressed profile. This distinction parallels the direct-effect versus stress-buffering contrast in the social support and conservation-of-resources literatures (Cohen & Wills, 1985; Hobfoll, 1989; Hobfoll et al., 2018) but is tailored to discrete profile transitions. Person-centered longitudinal research on motivational configurations similarly tests whether profiles and transitions carry distinct implications for adjustment (Howard et al., 2016, 2020).

Study 1: The Structure of Midlife Personality Profiles and Their Future-of-Work Skill Criteria

Study 1 examined whether latent personality profiles could be identified in a national adult panel, whether profile membership was associated with future-of-work skill criteria (Table A1), and whether profile membership improved prediction beyond continuous traits. Following recommendations for person-centered personality research (Meyer & Morin, 2016; Spurk et al., 2020), incremental validity was evaluated before interpreting profile-outcome associations. Study 1 also reports profile enumeration and cross-implementation validation of the

measurement model for Study 2. Hypotheses were not specified in Study 1 because they were developed after the LPA solution and profile labels were fixed (Kanitz et al., 2024; Kleshinski et al., 2024; Zhang et al., 2024).

Research Questions

RQ1. Are there quantitatively and qualitatively distinct latent personality profiles based on joint standing on Big Five and Agency indicators in the Midlife in the United States (MIDUS) national panel?

RQ2. Does independent re-enumeration recover the same number of profiles and the same core prototypes, and, do profile to skill-criterion relationships replicate in the independent MIDUS Refresher sample?

RQ3. Does profile membership improve prediction of skill criteria beyond continuous trait scores?

Method

Transparency and Openness

This manuscript reports a secondary analysis of publicly available MIDUS data obtained from the MIDUS data repository via the MIDUS Colectica portal and cross-referenced to Inter-university Consortium for Political and Social Research (ICPSR) study deposits (see Appendix B for data-access disclosure). Analysis code and derived output tables used to generate this draft are available from the lead author upon reasonable request; the underlying MIDUS data are publicly available (see Appendix B). MIDUS restricted-use files were not redistributed. The study used only de-identified public-use data and involved no new data collection from human participants. All analyses were conducted in R 4.5.2 (lavaan, LMest, and mclust; Scrucca et al., 2016) and Mplus 8.8.

Participants and Procedure

The study design was a nonexperimental, observational, longitudinal secondary analysis of an archival national panel. No conditions were manipulated or assigned, and all variables were naturally observed. Data were drawn from the MIDUS core panel M1 (1995-96), M2 (2004-06), and M3 (2013-14) (N = 7,108 at baseline) and the independent MIDUS Refresher (2011-14; N = 3,577). The MIDUS Refresher was used strictly as an external validation sample (Brim et al., 2004; Radler & Ryff, 2010). No participants were excluded from the core panel for missing work-role items. Employment status was coded as a subsample flag rather than a listwise deletion rule. Primary profile enumeration and incremental-validity checks used respondents with complete MIDUS 2 personality indicators (N = 3,970). Study 1 cross-sectional characterization and skill outcomes used MIDUS 2 as the primary assessment wave. Analyses of work-related outcomes used respondents who reported working or being self-employed at the relevant wave (N = 2,493 with complete personality data at M2), with working-only sensitivity checks. About two-thirds of employed respondents reported supervising others, permitting leader/manager-relevant description. Joint latent Markov models used all participants with usable personality data at consecutive waves (three-wave complete personality data: N = 2,615).

Sample size was determined by the archival design rather than by a priori power analysis. Analytic samples for each analysis family comprised all available cases meeting the completeness rules described above, and no cases were collected or excluded to reach a target sample size. The resulting samples are large for profile enumeration and full-panel models but smaller for pathway contrasts among Distressed movers, where precision is correspondingly lower (e.g., 86 modal Distressed respondents at MIDUS 2 for the MIDUS 2 to MIDUS 3

interval). Confidence intervals are reported for primary estimates so that precision can be evaluated directly.

Missing data were handled by analysis family. The joint latent Markov model was estimated with full information maximum likelihood (FIML) on all participants with usable personality data at consecutive waves (N = 6,438). For the regression-based transition, change, and outcome models, missing data were handled by complete-case (listwise) deletion on the variables entering each model; because the available sample therefore varies from model to model. The missing-data mechanism was not formally tested, and differential attrition was not modeled. Participant flow proceeded from the baseline panel (n = 7,108) to complete MIDUS 2 personality indicators (n = 3,970), the employed-at-MIDUS 2 subsample (n = 2,493), complete three-wave personality data (n = 2,615), baseline-Distressed respondents in leaving-versus-staying models (ns = 1,061-1,086 across predictor sets), and movers to Resilient or Reserved in direction models (n = 898).

Sample Characteristics

Respondents in the primary Study 1 sample (complete MIDUS 2 personality indicators; n = 3,970) averaged 56.11 years of age (SD = 12.32), and 55.5% were female. Mean education corresponded to some college without a degree (M = 7.26, SD = 2.54, on the 12-category MIDUS scale of highest level completed, where 5 = high school graduate, 7 = three or more years of college without a degree, and 9 = bachelor's degree). Race/ethnicity was 91.8% White, 3.6% Black, and 4.2% other. Table 1 reports descriptive statistics, internal consistencies, and correlations among the Midlife Development Inventory (MIDI) personality composites and the Study 2 outcomes.

Measures

Unless noted, scales were administered at every core wave. Composite scores follow MIDUS-project scoring rules, and analyses used the MIDUS-constructed composites rather than item-level responses. Internal consistency estimates (Cronbach's α) are therefore those reported in the MIDUS 2 psychometric documentation for the abbreviated scales used here (Ryff et al., 2007; see also Ryff & Keyes, 1995; Lachman & Weaver, 1998; Mroczek & Kolarz, 1998).

Personality Indicators. Six constructs (Neuroticism, Extraversion, Openness, Agreeableness, Conscientiousness, and Agency) were measured with the MIDI adjective ratings (Lachman & Weaver, 1997). Respondents rated how well each adjective described them on a 4-point scale (1 = not at all to 4 = a lot). Thirty adjectives were administered at M1 and 31 at subsequent waves (4-5 items per construct; e.g., moody, outgoing, creative, caring, organized, and self-confident). Profile indicators were ESEM factor scores rather than unit-weighted scale composites (see Analytic Approach). The MIDI was designed to assess the five-factor model together with a standalone Agency dimension, a six factor model previously shown to demonstrate adequate psychometric properties (Joshanloo, 2018; Zimprich et al., 2012).

Future-of-Work Skill and Individual Resources. Individual resources were mapped a priori to the future-of-work skill criteria (Table A1). Personal mastery (four items; e.g., "I can do just about anything I set my mind to"; α = .73) and perceived constraints (eight items, reverse-keyed; e.g., "I have little control over the things that happen to me"; α = .86) were used as measures of resilience and agency (Lachman & Weaver, 1998). Positive reappraisal (four items; e.g., "When I am faced with a bad situation, it helps to find a different way of looking at things"; α = .78) measured adaptability. Generativity was measured with the six-item Loyola scale (e.g., "I try to pass along knowledge I have gained through my experiences"; range 6-24; α = .85; McAdams & de St. Aubin, 1992). Episodic memory (e.g., immediate and delayed word-list

recall) and executive functioning (e.g., backward counting and category fluency) came from the Brief Test of Adult Cognition by Telephone at MIDUS 2 (Tun & Lachman, 2006). Karasek (1979) job-characteristics scales measured decision authority (six items; e.g., "How often do you have a say in decisions about your work?"; range 6-30; α = .87) and psychological demands (five items; e.g., "How often do you have too many demands made on you?"; range 5-25; α = .73) among employed respondents. Personal income and occupational prestige (GSS prestige scores from Census occupation codes) represented economic standing.

Subjective Well-Being. At MIDUS 2, four subjective well-being indicators were added: a single-item work-situation satisfaction rating ("Rate your current work situation"; 0-10), a single-item life-satisfaction rating ("Using a 0-10 scale where 0 means the worst possible life overall and 10 means the best possible life overall, how would you rate your life overall at this time?"; 0-10), and positive and negative affect scales (six items each, 1-5; e.g., enthusiastic and interested [positive]; upset and distressed [negative]; α = .90 and .85, respectively; Mroczek & Kolarz, 1998). The single-item job-satisfaction measure has estimated minimum reliability of .67-.70 against multi-item benchmarks (Wanous & Reichers, 1996; Wanous et al., 1997).

Covariates. Age (years), sex (female = 1), and education (MIDUS 12-category scale of highest level completed, 1 = no school/some grade school to 12 = doctoral or professional degree) were included as covariates in profile-outcome models and sensitivity checks. Employment status (working or self-employed = 1) defined subsamples for job-context outcomes.

Analytic Approach

Per-wave exploratory structural equation models (ESEM; Asparouhov & Muthén, 2009; Marsh et al., 2014) fit the MIDI item set acceptably (CFI = .93-.94, RMSEA = .05, SRMR = .02)

where independent-clusters CFA did not (CFI = .75-.78). Longitudinal invariance held at the metric level for all six constructs. Neuroticism and Conscientiousness required freeing one intercept each for partial scalar invariance, following established partial-invariance procedures (Byrne et al., 1989; Putnick & Bornstein, 2016; Vandenberg & Lance, 2000), after which longitudinally anchored factor scores (M1 metric) were extracted. The pattern of acceptable ESEM fit where independent-clusters CFA fails replicates prior psychometric evaluation of the MIDI adjective items (Joshanloo, 2018), and the metric-level invariance of the FFM constructs is consistent with evidence that these items support strict measurement invariance across age (Zimprich et al., 2012).

Analyses proceeded in four stages following person-centered best-practice recommendations (Meyer & Morin, 2016; Spurk et al., 2020). First, incremental validity was tested (shape versus level separation and out-of-fold prediction) using Gaussian mixture enumeration on MIDUS 2 factor scores. Second, joint latent Markov models (k = 2-5 latent classes) were estimated by maximum likelihood with multistart search and validated across two independently developed maximum-likelihood implementations (LMest, Bartolucci et al., 2017; Mplus 8.8, Muthén & Muthén, 2017; Nylund et al., 2007). Third, the mean personality configuration (profile centroid) of each latent class was characterized. Incremental-validity analyses on the cross-sectional MIDUS 2 enumeration used bias-adjusted BCH distal-outcome tests (Asparouhov & Muthén, 2014; Bakk & Vermunt, 2016; Bolck et al., 2004). Fourth, sensitivity analyses were conducted (split-half reproduction, covariance-structure constraints, Refresher replication, and a five-indicator refit omitting Agency). future-of-work skill-criterion and well-being associations for the k = 4 model are reported as covariate-adjusted means by modal profile assignment (controlling age, sex, and education), with omnibus profile F tests and

Tukey-adjusted pairwise comparisons. Because modal assignment entails some classification error at entropy = .71, pairwise letter groupings are interpreted descriptively.

Results

Study 1 results are organized in four sections: incremental validity of profile membership, latent profile enumeration and model selection, latent class profile characterization, and profile-outcome associations with the future-of-work skills.

Incremental Validity of Profile Membership

The incremental-validity test used a cross-sectional Gaussian mixture enumeration of MIDUS 2 factor scores, estimated separately from the joint longitudinal latent Markov model that provides the four-class core structure reported in the next section. Within split halves, this cross-sectional enumeration selected a two-class solution (see Table S1 in the online supplemental materials), and profile membership added essentially no out-of-fold predictive increment over the six continuous traits for any criterion (ΔR^2 between -.002 and +.007; e.g., sense of control: traits $R^2 = .358$ vs. traits + class = .358). The shape-versus-level test showed that class separation was predominantly elevational, that is, quantitative differences in overall level rather than qualitative differences in configural shape (Morin & Marsh, 2015). A supplementary dimensional benchmark reinforced this result. The recovery-versus-disengagement destination contrast (reported with the Study 2 transition analyses) added no more than .006 to R^2 for the psychosocial criteria once continuous within-person change on the personality elevation axis was controlled. Incremental variance was nonsignificant for every criterion (all $p \geq .05$), answering RQ3. Profiles do not add incremental prediction beyond the dimensions in these data. It does not, however, negate the four-class longitudinal solution identified by the joint three-wave model. Consistent with person-centered practice in which

configural types describe subpopulations and developmental structure while continuous dimensions describe predictive variance (He et al., 2023; Kanitz et al., 2024; Zhang et al., 2024), all subsequent profile findings are advanced as descriptive and developmental, not as predictive superiority claims.

Latent Profile Enumeration and Model Selection

An initial three-class joint latent Markov solution recovered the classic ROU prototypes and passed an extensive set of robustness checks (split-half Tucker's φ >= .997 [Lorenzo-Seva & ten Berge, 2006]; equal-covariance sensitivity; Refresher replication of skill-criterion associations). A more thorough multistart search, however, showed this solution to be a local optimum, a well-documented risk in finite-mixture and latent-transition estimation (Hipp & Bauer, 2006; McLachlan & Peel, 2000). With additional random starts, estimation reached a deeper log-likelihood at k = 3 that both independently developed implementations reproduced. A full multistart re-enumeration (k = 2-5; Table 2) found monotonically decreasing BIC through k = 5, but the k = 5 solution failed a 100-start stability audit (its best likelihood was never reproduced and its class structure shifted across deep starts), whereas the k = 4 solution was recovered repeatedly with an identical class structure and restored every class of the published prototype set. The k = 4 model was therefore adopted as the core structure and confirmed by an independent maximum-likelihood implementation (Mplus 8.8; log-likelihood = -84,834.97; BIC = 170,301; entropy = .71). Latent class profile means matched across the two implementations essentially exactly, and modal classifications agreed 99.1%-99.7% across waves (M = 99.5%). This enumeration correction is reported in full because single-start estimation is the default in several latent Markov packages. All primary Study 2 analyses used the corrected k = 4 model, and references to an initial three-class solution appear only in this enumeration correction.

Selecting k = 4 over the monotonically-lower-BIC k = 5 solution on stability and class-distinctiveness grounds follows established person-centered practice, in which the number of profiles is judged on interpretability, class distinctiveness, and replicability rather than on fit indices alone when those indices continue to improve with each added class (Kanitz et al., 2024; Morin, 2016; Spurk et al., 2020).

Latent Class Profile Characterization

The four profiles (Table 3; Figure 1, Panel A) were labeled with parallel single-word descriptors consistent with person-centered reporting conventions (Meyer & Morin, 2016; Spurk et al., 2020): Resilient, Distressed, Reserved, and Antagonistic. Labels were assigned from the stable k = 4 mean structure (Distressed = highest Neuroticism; Antagonistic = lowest Agreeableness; Resilient = lower Neuroticism among the remaining classes; Reserved = the non-extreme withdrawn class).

With respect to prior FFM typologies, the four profiles map onto recurrent templates with deliberate deviations. The Resilient profile (36% at baseline; lowest Neuroticism, high Agreeableness and Conscientiousness) corresponds to the resilient and role-model configurations described by Block and Block (1980), Robins et al. (1996), Asendorpf et al. (2001), and Gerlach et al. (2018). The Distressed profile (30%; highest Neuroticism with preserved Agreeableness and Extraversion) aligns with the internalizing distress pole of the ROU framework (Robins et al., 1996; Donnellan & Robins, 2010) but departs from classic overcontrolled prototypes that pair high Neuroticism with social inhibition. Distress here co-occurs with intact communion, the warm, other-oriented interpersonal orientation indexed by Agreeableness in the MIDI and paired conceptually with Agency in the agency-communion framework (Bakan, 1966; Wiggins, 1991).

This combination distinguishes the class from Type D (Denollet, 2005) and from socially withdrawn distress.

The Antagonistic profile (5%; Agreeableness 2.2 SD below the mean, low Extraversion) recovers the low-communion extreme that appears as self-centered or antagonistic types when mixture models permit more than three classes (Gerlach et al., 2018; Isler et al., 2017; Lynam & Miller, 2019). In the previous three-class solution of the joint model, this pole was merged with a larger withdrawn class into a single detached-antagonistic state. The k = 4 solution separates the extreme low-communion core from the broader withdrawn majority.

That majority was labeled Reserved (29%; modestly low Extraversion and Agreeableness, near-average Neuroticism and Conscientiousness). The name is parallel to the other profile labels and denotes social reserve (interpersonal caution and low engagement) rather than a claim of exact identity with the Reserved class in Gerlach et al. (2018), whose prototype shows above-average Agreeableness and Conscientiousness. In MIDUS, the class instead corresponds to the large, non-extreme withdrawn configuration that remains after the antagonistic core is split off. The configuration shows low communion without the 5% extreme and modest distinctiveness on all traits, and it is the secondary destination for transitions out of long-standing antagonism. A sensitivity analysis dropping Agency (Big Five indicators only; 25 random starts) recovered the same four profiles with near-identical means and 97.8%-98.4% per-wave modal agreement with the six-indicator solution, indicating that the solution is not an artifact of including Agency.

Profile-Outcome Associations With Future-of-Work Skills

Profiles showed differentiated associations with the future-of-work skill criteria (Table 4; Figure 1, Panel C). Resilient members reported the highest psychosocial skill criteria (sense of

control, reappraisal, generativity, purpose, and mastery; all omnibus $p < .001$). Antagonistic members reported the highest personal income, occupational prestige, and BTACT analytic composite, with Reserved second, consistent with the agreeableness penalty in earnings (Judge et al., 2012; Nyhus & Pons, 2005). Psychosocial skill criteria and income, prestige, and analytic performance therefore represented distinct axes of profile variation.

Well-being outcomes also differed between profiles (Table 5). Antagonistic members reported life satisfaction comparable to Distressed members (7.35 vs. 7.36; Resilient = 7.96; omnibus $p < .001$) and positive affect well below Resilient (Antagonistic = 3.12, Distressed = 3.03, Resilient = 3.56), despite higher income and prestige. Job satisfaction followed the same ordering (Antagonistic 7.22 vs. Resilient 7.64). This pattern is consistent with SDT goal-contents research (Kasser & Ryan, 1996; Niemiec et al., 2009). A profile-by-income interaction showed income was unrelated to life satisfaction within the Antagonistic profile ($b = -0.02$, $p > .05$) but positively related within the Distressed profile ($b = +0.35$, $p < .001$), relative to the Resilient slope.

Study 1 Discussion

Study 1 addressed three research questions about profile structure, replication, and incremental validity. RQ1 was answered affirmatively, with an important qualification about the nature of the separation. A four-profile solution (Resilient, Distressed, Reserved, and Antagonistic) was recovered in a national panel, was robust to multistart estimation and to re-estimation in an independent maximum-likelihood implementation, and aligned with recurrent FFM latent profile templates while splitting the former detached mass into Reserved and Antagonistic constituents (Gerlach et al., 2018; Robins et al., 1996; Yin et al., 2021). Profile separation was predominantly elevational, reflecting quantitative differences in overall level

more than qualitative differences in configural shape (Morin & Marsh, 2015); the principal configural marker is the Distressed profile's elevated Neuroticism relative to its preserved communion. Regarding RQ2, independent multistart re-enumeration in a separate sample recovered the four-profile granularity on BIC and reproduced the dominant Resilient prototype (Tucker's $\varphi = .81$), and the external skill-criterion patterns replicated in the independent Refresher sample. The Distressed configuration, however, did not re-emerge as a standalone cross-sectional cluster under independent re-enumeration in the Refresher and, on the same cross-sectional test, was not recovered in the core MIDUS 2 sample either; it is defined instead by its position in the joint three-wave transition model.

The Distressed profile is therefore best understood as a longitudinally defined status, identified from its transition dynamics, rather than as a configuration that recurs in any single cross-section. This is consistent with latent transition research in which statuses are transient and defined by movement over time (Collins & Lanza, 2010; Kleshinski et al., 2024; Slaughter et al., 2021). Specifically regarding RQ3, profile membership did not improve prediction beyond continuous traits, so subsequent profile findings are interpreted descriptively and developmentally rather than as evidence of predictive superiority. This is in line with previous person-centered work in which configural types are descriptive of subpopulations while continuous dimensions are predictive (He et al., 2023; Kanitz et al., 2024; Zhang et al., 2024).

Study 1 also documented a skill-criterion difference between profiles at MIDUS 2. Psychosocial skill criteria were highest in the Resilient profile whereas analytic performance (as well as personal income and occupational prestige) were highest in the Antagonistic profile. While income and prestige were higher in the Antagonistic profile this did not cohere with psychosocial well-being as job and life satisfaction were lowest in the Antagonistic and

Distressed profiles. In goal-contents terms (Kasser & Ryan, 1996; Niemiec et al., 2009), the profile contrast separates intrinsic growth-related resources, concentrated in the Resilient profile, from extrinsic rewards (income and prestige), concentrated in the Antagonistic profile, with cognitive performance alongside the extrinsic-economic criteria without being a motivational outcome itself.

Study 1 is nevertheless limited in the ways that are typical of cross-sectional person-centered discovery studies (Kleshinski et al., 2024). It cannot establish whether profile membership is stable over time, whether transitions are asymmetric, whether BPN predicts recovery-oriented versus disengagement-oriented transition direction, or whether daily-ecology corroborates profile interpretation. With these limits in mind, and staying with the exploratory-to-confirmatory sequence (Kanitz et al., 2024; Kleshinski et al., 2024; Zhang et al., 2024), hypotheses were deferred to Study 2 after the four-class solution and profile labels were fixed.

Study 2: Profile Transitions, BPN, and the Daily Ecology of Recovery Versus Disengagement

Study 2 extends the four-class measurement model selected in Study 1 to estimate profile transitions across approximately twenty years. The four-class structure was replicated to confirm Study 1's latent profile prototype (Asendorpf et al., 2001; Gerlach et al., 2018). Following Chawla et al. (2020), event-sampled daily diary data provide elaboration and corroboration of profile interpretation at the daily level (Creswell & Plano Clark, 2018; Johnson & Christensen, 2017). Follow-up analyses address job context and additional diary contrasts.

Study 1 identified a Distressed configuration, a Resilient recovery target, and a Reserved disengagement pathway among Distressed movers. Following the two-study logic used when hypothesis tests depend on an initial latent profile solution (Gabriel et al., 2015; Howard et al.,

2020), Study 2 hypotheses were developed from that structure and from the skill-criterion relationships observed in Study 1. Profiles are treated as cross-trait patterns at each wave. Transitions between profiles reflect the reallocation of individuals across these configurations over time, that is, genuine personality change across adulthood (Roberts & Mroczek, 2008).

Hypotheses

H1 and H2 concern the structure of change and specifically examine whether the Distressed profile becomes increasingly sparse across two decades and whether those who leave this profile follow distinct transition pathways. H3 concerns the substance of change and examines whether movement in profile standing is accompanied by gains in socioemotional skill criteria versus cognitive skill criteria. H4 and H5 test the SDT contention that BPN satisfaction governs the quality of personality integration under strain. H4 predicts the direction that movers take, and H5 predicts the unique relation of each need to a matched skill criterion. H6 extends the aforementioned hypotheses to a performance criterion, health-related lost productive time, connecting profile position and transition direction to a standard organizational outcome. SDT supplies a unifying theoretical mechanism. Need satisfaction is theorized to foster integration and re-engagement, whereas need frustration is theorized to foster defensiveness and withdrawal (Deci & Ryan, 2000; Ryan & Deci, 2017; Vansteenkiste & Ryan, 2013).

Meta-analytic and longitudinal evidence documents normative declines in Neuroticism from early to late adulthood, alongside gains in the traits that anchor adaptive functioning (Mroczek & Spiro, 2007; Roberts et al., 2006). SDT supplies a mechanism for this maturational drift. On the organismic-integration view, adults tend toward greater personality integration with age when environments are not chronically need-thwarting, and goal content shifts toward intrinsic, generative strivings across the life span (Ryan & Deci, 2017; Sheldon & Kasser, 2001).

The Distressed profile combines elevated Neuroticism with preserved communion (warmth toward others). Because greater personality integration tends to resolve exactly this combination, the profile should become less common as integration increases with age (Roberts et al., 2006; Ryan & Deci, 2017). Person-centered evidence is consistent with this expectation in that type proportions shift across development rather than remaining fixed (Asendorpf & van Aken, 2003; Howard et al., 2020).

H1. The proportion of adults classified in the Distressed profile will decrease from MIDUS 1 to MIDUS 3.

Multifinality, the principle that the same starting configuration can yield qualitatively distinct pathways with different correlates, is a recurrent theme in developmental typology research (Bergman & Magnusson, 1997; Cicchetti & Rogosch, 1996; von Eye & Bergman, 2000). SDT provides a framework to further clarify this principle via a directional prediction. Namely, its dual-process account holds that need satisfaction strengthens the inner resources that support proactivity and re-engagement, whereas need frustration promotes defensiveness, withdrawal of investment, and the pursuit of need substitutes (Ryan & Deci, 2017; Vansteenkiste & Ryan, 2013). Individuals leaving the Distressed profile should accordingly be distinguished by at least two qualitatively different pathways. One pathway involves recovery, in which individuals demonstrate increased engagement and communal functioning (movement to Resilient). A second pathway involves disengagement, wherein distress abates through withdrawal rather than re-engagement (movement to Reserved). Person-centered recovery research finds this distinction, with daily experience configurations resolving into divergent adjustment patterns rather than a single recovery trajectory (Chawla et al., 2020). If these pathways are genuinely distinct rather than arbitrary partitions of one process, they should differ

in their skill-criterion correlates. Recovery should preserve or restore socioemotional skill criteria, whereas disengagement should erode them.

H2. Adults who leave the Distressed profile will follow distinct transition pathways that differ in their skill-criterion correlates. Recovery-oriented transitions (to Resilient) will show higher sense of control, reappraisal, and generativity than disengagement-oriented transitions (to Reserved).

The pathways should also differ in what changes when profile standing improves. Personality-linked self-beliefs and socioemotional resources show greater within-person plasticity across adulthood than fluid cognitive ability (Deary et al., 2000; Lachman & Weaver, 1998; Roberts & Mroczek, 2008). SDT posits that this plasticity is driven by need-satisfying experiences that strengthen the self-regulatory and relational resources through which integration is expressed. These self-regulatory and relational resources include mastery, reappraisal, and generativity. Fluid cognitive performance is not a motivational outcome and was expected to be comparatively inert over each decade-long interval (Deary et al., 2000; Salthouse, 2009). The differential prediction also serves a discriminant function. If profile improvement reflected measurement artifact, such as regression to the mean or shifting response style, rather than genuine developmental reorganization, gains would not concentrate selectively in the socioemotional criteria.

H3. Within-person improvement in profile standing (movement toward a more adaptive, higher-elevation profile configuration) will positively predict gains in socioemotional skill criteria across each longitudinal interval.

Self-determination theory holds that satisfaction of the needs for autonomy, competence, and relatedness enhances the quality of personality integration under strain (Deci & Ryan, 2000;

Ryan & Deci, 2017). Meta-analytic evidence supports this premise in the work domain. Each of the three needs uniquely predicts growth, internalization, and well-being (Van den Broeck et al., 2016), and need-supportive work design predicts motivation and adjustment (Deci et al., 2017; Gagné et al., 2022). Purpose in life, which is conceptually adjacent to SDT rather than a basic need, is an antecedent of integration and was treated as a supplementary predictor (Park, 2010; Steger et al., 2012). Person-centered transition research frames such antecedents as shifting the probability of membership in adaptive versus maladaptive configurations (Gabriel et al., 2015; Howard et al., 2016, 2020). Applied to the direction of transition, the dual-process logic developed for H2 implies that movers whose needs are comparatively satisfied hold the resources that re-engagement requires and should favor the recovery pathway, whereas movers whose needs are comparatively frustrated should leave distress by withdrawal, favoring the disengagement pathway (Vansteenkiste & Ryan, 2013).

H4. Among adults who leave the Distressed profile, autonomy, competence, relatedness, and supplementary purpose in life will be positively associated with recovery-oriented (Resilient) versus disengagement-oriented (Reserved) transition direction.

SDT further specifies that the three needs are functionally distinct rather than interchangeable. Autonomy supports volitional self-regulation, competence supports the experience of effectance (feeling capable and producing intended effects; White, 1959) and growth, and relatedness supports interpersonal integration (Ryan & Deci, 2017). Meta-analytic work reinforces the distinction, showing that the needs predict outcomes unevenly (Van den Broeck et al., 2016), and organizational need-support interventions produce domain-specific rather than undifferentiated benefits (Gagné et al., 2022). Matching each need to the skill criterion that most directly expresses its function yields three targeted predictions. First,

competence, the experience of effectance (White, 1959), should lead to generativity, the outward transmission of skill and knowledge, because an established sense of one's own capability is what frees a person to invest in developing others (McAdams & de St. Aubin, 1992). Second, relatedness should support positive reappraisal, because secure connection supplies the interpersonal scaffolding on which adaptive reframing of adversity is built, and social support is a documented antecedent of reappraisal (Marroquin, 2011). Third, autonomy, the experience of volition, should consolidate into a generalized sense of control (Lachman & Weaver, 1998). Each matched prediction was nonetheless tested against the full criterion space, allowing off-target associations to surface.

H5. Across the full panel, competence will positively predict generativity change, relatedness will positively predict reappraisal change, and autonomy will positively predict sense-of-control change.

The final hypothesis extends the SDT psychological framework to a performance consequence. Health-related lost productive time is a standard organizational outcome for distress and functional impairment (Stewart et al., 2003). In SDT terms, the Distressed configuration is the profile-level pattern of compromised integration, and need frustration is theorized to manifest in impaired daily functioning as well as ill-being (Vansteenkiste & Ryan, 2013). In work settings, need satisfaction predicts both wellness and performance (Deci et al., 2017). Detriments to performance should therefore be most pronounced in the Distressed profile. The same logic extends to the differentiation in individual transition pathway. The recovery pathway restores engaged functioning and should return productive capacity, whereas the disengagement pathway conserves resources through withdrawal without restoring engagement, so its reductions in impairment should be attenuated.

H6. Distressed profile membership will be associated with greater health-related productive impairment than membership in other profiles. Among adults who leave the Distressed profile, recovery-oriented transitions will show larger reductions in impairment than disengagement-oriented transitions.

Method

Study 2 uses the measurement model, profile labels, and analytic sample definitions established in Study 1; all additional measures, exclusions, and analysis plans are described below. Transparency and openness disclosures are provided in the Study 1 Method and apply to both studies.

Participants and Procedure

Study 2 is a secondary analysis of the same nationally representative longitudinal panel described in Study 1 (Bryman, 2012). Analyses used the MIDUS core panel, following respondents across approximately twenty years (M1-M3). Joint latent Markov transition models used all participants with usable personality data at consecutive waves. Transition-predictor models for organizational claims used the employed-at-M1 subsample (working or self-employed at baseline). Parallel models on the full panel provide robustness checks (Table 6). The three National Study of Daily Experiences (NSDE) subsamples supplied event-sampled daily-ecology data measured independently of the trait self-report battery.

Measures

Personality indicators and covariates were identical to Study 1. Additional Study 2 measures are described below.

Basic Psychological Needs. Each need was assessed with an established scale. Autonomy used the Ryff well-being autonomy subscale (three items; e.g., "I judge myself by

what I think is important"; α = .45; Ryff, 1989; Ryff & Keyes, 1995). Competence was assessed with the personal mastery scale (four items; e.g., "I can do just about anything I set my mind to"; α = .73; Lachman & Weaver, 1998). Relatedness used the Ryff positive relations with others subscale (three items; e.g., "People would describe me as a giving person, willing to share my time with others"; α = .63; Ryff, 1989). Purpose in life (three items; e.g., "Some people wander aimlessly through life, but I am not one of them"; α = .29; Ryff, 1989) was retained as a supplementary eudaimonic indicator rather than a basic SDT need. Decision authority (six items; e.g., "How often do you have a say in decisions about your work?"; α = .87) served as a contextual indicator of workplace autonomy support among employed respondents (Karasek, 1979). Negative work-to-family spillover (four items; e.g., stress at work carrying over into home life) was included as a work-strain covariate among employed respondents.

Health-related lost productive time. At every wave, respondents reported days in the past 30 completely unable to work or carry out normal activities (e.g., "About how many days were you totally unable to work or carry out your normal activities?") and days cut back because of health problems, with physical/mental attribution at later waves (Stewart et al., 2003). The LPT index equals unable days plus half of cutback days and was turned into a dollar cost metric by the human-capital method (Rice, 1967; Stewart et al., 2003) for economic costing.

Daily diary measures (NSDE). Three NSDE subsamples provided up to eight consecutive nightly telephone interviews: NSDE II (2004-09, concurrent with MIDUS 2), the MIDUS 3 Daily Diary (2017-19), and the Refresher Daily Diary (2012-14). Daily-ecology elaboration analyses in the primary text used NSDE II person-level aggregates classified under modal k = 4 membership at MIDUS 2. The Daily Inventory of Stressful Events (DISE; Almeida et al., 2002) yielded, for each day, expert-coded stressor occurrence and respondent ratings of severity and

appraised threat. Daily negative and positive affect scales were also administered. Person-level aggregates (at least three valid days) comprised stressor exposure (share of days with any stressor; e.g., argument with spouse), interpersonal-conflict exposure, mean severity, appraised threat to control-relevant domains (daily routine, future plans) versus self/social-evaluative domains, mean daily negative and positive affect, and affective reactivity (negative affect on stressor days minus stressor-free days; Almeida, 2005). The MIDUS 1 NSDE (1996-97) provided the same aggregates prior to the first transition interval, permitting prospective tests.

Analytic Approach

Study 2 analyses used the k = 4 joint latent Markov model selected in Study 1 (longitudinally anchored factor scores, M1 metric) and its estimated initial and transition probabilities, alongside modal k = 4 assignments for descriptive flows, subgroup analyses, and daily-ecology corroboration analyses at MIDUS 2.

Transition-likelihood and transition-direction tests were logistic regressions among baseline-Distressed respondents (leaving vs. staying; recovery vs. disengagement pathway among movers to Resilient or Reserved), adjusted for age, sex, and education, with all needs entered jointly to isolate unique contributions. Likelihood models test whether a need predicts leaving the Distressed profile. Direction models test, among movers, whether a need predicts recovery (Resilient) versus disengagement (Reserved). Likelihood models were estimated descriptively to contextualize H4 against direct-effect and buffering accounts (Cohen & Wills, 1985; Hobfoll, 1989). Need-skill-criterion tests regressed each follow-up criterion on its baseline value, the three needs jointly, and demographics (residualized change). Organizational transition predictors are reported primarily for the employed-at-M1 subsample. Full-sample estimates appear in Table 6 for comparison. Ninety-five percent confidence intervals accompany primary

estimates. Intervals were computed from estimated standard errors where the analysis output preserved them and were otherwise reconstructed as approximate Wald intervals from point estimates and exact p values, the method also used for Figure 2. For odds ratios near 1.00, reconstructed intervals are approximate and should be read as precision indicators rather than exact profile-likelihood bounds.

Each of the six confirmatory hypotheses was evaluated at conventional significance levels (i.e., $\alpha = .05$) and because these were a priori directional predictions rather than a single undifferentiated family of comparisons, no familywise (e.g., Bonferroni) correction was applied across them (cf. Vaziri et al., 2020). Exploratory analyses were likewise not adjusted for multiple comparisons and are interpreted descriptively.

Results

Study 2 results demonstrate latent profile and latent transition sequence using a person-centered longitudinal approach (Kleshinski et al., 2024; Slaughter et al., 2021; Vaziri et al., 2020). Hypotheses are reported in order: transition structure (H1), heterogeneous transition pathways (H2), within-person profile change and skill criteria (H3), SDT predictors of transition direction (H4), need-skill-criterion specificity (H5), and productive-time associations (H6), followed by daily-ecology elaboration and corroboration with event-sampled diary data.

Latent Profile Transitions

H1 was supported. Transitions were asymmetric and predominantly toward healthier configurations (Figure 1, Panel B; Table 7). The Distressed profile held 29.5% of the sample at MIDUS 1 and declined to 8.8% at MIDUS 2 and 7.2% at MIDUS 3 (model-based). Figure 1, Panel B and Table 7 present these model-based prevalences and the transition probabilities among them. Transitions out of the Distressed profile went predominantly to Resilient (transition

probability = .52) with a secondary flow to Reserved (.18). Entry into the Distressed profile was near zero from every origin (<= .02; two transition logits were empirically at singularity, i.e., structural zeros). The subgroup still Distressed at MIDUS 2 was consistent across all phases (.75 stayed to MIDUS 3). The Antagonistic profile persisted at a probability of approximately .5 per decade, and most remaining transitions were to Reserved. The Antagonistic profile was moderately persistent but essentially never converted to Distressed or directly to Resilient. The decline showed only a small association with baseline age: transition rates were 82%-89% across baseline-age tertiles (adjusted OR = 1.20 per SD of age), and movers' Neuroticism declined at roughly three times the normative sample-wide amount, so the decline is not purely attributable to maturational trait change (Roberts et al., 2006). The pattern is more consistent with episodic distress resolving over a long interval, leaving a small enduring subgroup.

Profile transition data suggest the most common pattern involved maintaining the same profile across each decade-long interval and that stability varied by configuration (Kleshinski et al., 2024; Slaughter et al., 2021; Vaziri et al., 2020). Resilient and Antagonistic profiles were the most stable, whereas Distressed membership was the least stable and showed the asymmetric decline that motivated Study 2 hypotheses. Among Distressed adults who moved, the primary transition was recovery to Resilient (transition probability = .52), consistent with normative Neuroticism decline in adulthood but larger in movers than in the sample-wide average. A secondary transition was disengagement to Reserved (.18), a pathway with distinct skill-criterion correlates in Study 2. Entry into the Distressed profile was near zero from every origin (<= .02), indicating that new distress onset was rare in this panel. Probabilities of extreme reconfiguration (for example, direct movement between the most contrasting configurations without intermediate recovery) were correspondingly low and should not be over-interpreted, paralleling the rare

extreme transitions reported in recent latent transition analyses (Kleshinski et al., 2024; Slaughter et al., 2021).

At the item level, the transition was carried disproportionately by the anxiety markers of the adjective battery. Among M1-Distressed adults, movers declined 0.79 and 0.73 raw points (4-point scale) more than stayers on nervous and worrying, versus 0.42 on moody and 0.16 on (low) calm, with parallel changes of the communal adjectives (sympathetic, caring, softhearted: 0.21-0.27). Recovery from the Distressed profile is, descriptively, an anxious-apprehension pattern rather than a general neuroticism story (supplementary descriptive decomposition; the MIDI carries no formal facet structure).

Heterogeneous Transition Pathways From the Distressed Profile

H2 was supported. Transitions out of the Distressed profile took two behaviorally distinct pathways (Tables 8 and 9), a pattern consistent with multifinality in person-centered development (Bergman & Magnusson, 1997; Cicchetti & Rogosch, 1996). Recovery (to Resilient) brought higher sense of control (+0.31 SD vs. stayers, 95% CI [.20, .42]) and reappraisal (+0.21 SD, 95% CI [.10, .32]) with generativity preserved. Disengagement (to Reserved) left control unchanged but cost generativity (-0.42 SD, 95% CI [-.57, -.27]) and reappraisal (-0.21 SD, 95% CI [-.37, -.06]). Transition likelihood was predicted by declining Neuroticism, low perceived constraints, and reappraisal. Negative work-to-family spillover was negatively associated with recovery-oriented transitions, and lower generativity was associated with disengagement-oriented transitions. The two-pathway structure reproduced inside the four-class core model, in which 72% of modal M1-Distressed movers landed in Resilient and 12% in Reserved, with under 2% entering the Antagonistic profile. Baseline-adjusted M2 income, occupational prestige, decision authority, and supervisory status did not differ between recovered

movers and stayers (all ps > .50; exploratory). Associations with health-related lost productive time are reported with H6 below.

Within-Person Profile Change and Skill Criteria

H3 was supported. Within-person gains in profile elevation predicted skill-criterion gains in both intervals (e.g., reappraisal β = .39, 95% CI [.36, .42], and .30, 95% CI [.26, .33]; sense of control β = .38, 95% CI [.35, .41], and .32, 95% CI [.29, .35]; all p < .001), whereas BTACT change was small by comparison (β = .04, 95% CI [.02, .07], p = .002), consistent with the theoretical expectation that analytic performance would show limited plasticity over each interval.

Basic Psychological Needs, Purpose, and Transition Direction

H4 received partial support. The three BPNs were not interchangeable (Table 6; Figure 2). Among full-panel movers (n = 898 to Resilient or Reserved), autonomy was marginally associated with recovery direction (OR = 1.20, 95% CI [0.99, 1.45], p = .059), competence did not (OR = 1.01, 95% CI [0.88, 1.16], p = .89), relatedness was directionally positive but not significant (OR = 1.15, 95% CI [0.95, 1.39], p = .15), and purpose predicted direction (OR = 1.23, 95% CI [1.01, 1.50], p = .039). Results were unchanged in the employed-at-M1 subsample (n = 657 movers). In that subsample, purpose (OR = 1.25, 95% CI [1.01, 1.55], p = .043) and relatedness (OR = 1.22, 95% CI [0.98, 1.51], p = .069) were associated with direction, autonomy remained marginal (OR = 1.16, 95% CI [0.93, 1.45], p = .19), and competence remained null (OR = 1.01, 95% CI [0.86, 1.18], p = .90).

Descriptive likelihood models provided a useful contrast. In the full panel, no basic need or purpose predicted transition likelihood (autonomy OR = 1.07, p = .45; competence OR = 1.09, p = .30; relatedness OR = 1.09, p = .32; purpose OR = 0.88, p = .17), and the same pattern held

among employed-at-M1 respondents. Among employed movers, decision authority showed a parallel pattern in an exploratory follow-up (direction OR = 1.17, p = .17; likelihood OR = 1.04, p = .69), paralleling PWB autonomy rather than reversing it. Buffering effects by purpose also failed (1 of 8 interactions nominally significant and sign-flipping across intervals), whereas purpose main effects on skill-criterion change were consistently positive, a pattern consistent with a direct effect rather than buffering moderation (Cohen & Wills, 1985).

Need-Skill-Association Specificity Across the Full Criterion Space

H5 was partially supported. Extending to the whole sample (44 full-sample residualized-change tests in a joint four-predictor model, Table 10; 172 tests including the k = 3 stratum battery), two of the three preplanned need-criterion links replicated across both decade-long intervals. Autonomy uniquely predicted sense-of-control change (β = .04, 95% CI [.01, .07], in both intervals) and relatedness uniquely predicted reappraisal change (β = .05, 95% CI [.02, .08], and .06, 95% CI [.03, .09]). The third link was interval-specific. Competence predicted generativity change from MIDUS 1 to MIDUS 2 (β = .03, 95% CI [.003, .06], p = .027) but not from MIDUS 2 to MIDUS 3. Beyond the preplanned links, relatedness was the broadest prosocial predictor (generativity β = .06 and .07 across the two intervals), competence showed the single largest unique effect (sense-of-control change, β = .08, MIDUS 1 to MIDUS 2), and supplementary purpose showed an agentic-economic pattern (sense of control and personal income in both intervals; occupational prestige from MIDUS 1 to MIDUS 2). Significant unique effects were small per decade (β = .03-.08) but directionally uniform. Unexpectedly, purpose predicted BTACT change in the M2-M3 interval (full sample β = .030, p = .026; subgroup remaining Distressed, β = .109, p = .025), the only SDT antecedent of the analytic criterion in this project.

Lost Productive Time by Transition Pathway

H6 was supported. MIDUS administers the standard health-related lost-productive-time items at every wave. The items assess days in the past 30 the respondent was completely unable to work or carry out normal activities, and additional days they had to cut back, because of physical or mental health. Following the lost-productive-time convention (unable days plus half of cutback days; Stewart et al., 2003), the profiles showed a strong impairment gradient at both waves bracketing the first transition interval (adjusted days per month at MIDUS 2: Distressed 3.09, Reserved 1.66, Antagonistic 1.35, Resilient 1.24; $p < .001$). Transition pathway then predicted the change. Adjusting for baseline impairment and demographics, recovered movers logged 1.34 fewer impaired days per month than stayers at MIDUS 2 (95% CI [-2.12, -0.55], $p = .001$; adjusted means 1.53 vs. 2.85), whereas disengaged movers' reduction was half the size and not significant (-0.76, 95% CI [-1.80, 0.27], $p = .15$). Recovery restored productive capacity; the disengagement pathway's reduction was smaller and nonsignificant. Valued by the human-capital method (impaired days x 12 x daily wage from personal income; Rice, 1967; Stewart et al., 2003), Distressed workers lost roughly $5,600 per year in productive time against $1,800 for Resilient workers (2004-06 dollars), and leaving was worth approximately $3,000 per worker-year relative to staying (recovered -$3,049, 95% CI [-$5,038, -$1,060], $p = .003$; disengaged -$3,209, 95% CI [-$5,690, -$728], $p = .01$; the disengaged dollar figure reflects wage composition as much as day counts). For illustration, given the non-experimental longitudinal panel design, with roughly 30% of the panel Distressed at baseline and 72% of movers recovering, each 1,000 midlife workers carry on the order of $650,000 per year in recoverable productive time. These dollar estimates are illustrative rather than definitive; interpretive and boundary caveats are addressed in the Limitations.

Daily-Ecology Elaboration and Corroboration

To elaborate and corroborate profile interpretation with an independent quantitative modality, modal k = 4 profile membership was examined for prediction of daily-ecology indicators measured in event-sampled diaries (Table 11). In NSDE II, concurrent with MIDUS 2 (n = 1,106 linked person-level aggregates with at least three valid diary days), profiles differed on six of seven daily-ecology measures (all $p < .03$ except affective reactivity, $p = .29$). Distressed adults showed the highest daily negative affect and severity. Resilient adults showed the lowest negative affect and highest positive affect. Reserved adults showed muted disengagement (low interpersonal-conflict exposure with depressed positive affect). Antagonistic adults showed the highest stressor-day share and evaluative threat with only intermediate negative affect, reflecting high stressor exposure without correspondingly high distress, consistent with the four-class measurement model.

Replication in the MIDUS 3 diary subsample (n = 463; modal k = 4 assignment at MIDUS 3) preserved the core affective gradient: Distressed adults again showed the highest negative affect (.292 vs. .138 Resilient, $p < .001$) and lowest positive affect (2.40 vs. 2.84, $p < .001$), with significant omnibus tests also for severity and evaluative threat. The Refresher diary subsample (n = 745; k = 4 centroid classification on anchored factor scores) reproduced the same affective gradient across all four profiles: Distressed adults again showed the highest daily negative affect (.385 vs. .146 Resilient, $p < .001$) and the highest evaluative-threat appraisals (.514 vs. .351 Resilient, $p = .004$), whereas Resilient adults showed the highest positive affect (2.88 vs. 2.32 Distressed, $p < .001$). Affective reactivity was nonsignificant in the NSDE II ($p = .29$) and MIDUS 3 ($p = .075$) diaries but significant in the Refresher, where Distressed adults showed the largest reactivity ($p < .001$).

Transition pathways determined distinct daily appraisals and affective responses. Among M1-Distressed adults with M2 diaries, recovered movers faced the same stressor exposure as stayers but appraised events as less severe, reacted less, and uniquely showed restored positive affect (2.71 vs. 2.45, $p = .002$). Disengaged movers reduced negative affect without any positive-affect recovery, a daily-level pattern paralleling their generativity loss. Prospectively, the MIDUS 1 diary (which precedes the first transition interval) showed that daily interpersonal-conflict exposure predicted remaining Distressed a decade later (OR = 0.64 per SD for leaving, $p = .001$), whereas appraised severity and threat did not (ORs near 1.0). Chronic distress was forecast by the frequency of daily conflict exposure rather than by appraised severity or threat.

Study 2 Discussion

Study 2 extends Study 1 by modeling latent profile transitions across two decades. Profile prevalence shifted asymmetrically toward lower Distressed membership, with little new entry into distress, a pattern more consistent with episodic strain resolving than with accumulating workforce distress. Transitions out of the Distressed profile followed two interpretable pathways. Recovery preserved the generative and prosocial skills emphasized in future-of-work core- and rising-skills lists, whereas disengagement did not. H4 received partial support. Among movers, purpose predicted recovery-oriented direction, with autonomy and relatedness showing parallel marginal trends. Descriptive likelihood models showed no comparable associations with whether Distressed adults left at all. Event-sampled daily diary data elaborated and corroborated these patterns at the daily level. Recovered movers reported lower appraised severity and higher positive affect than stayers, whereas disengaged movers showed reduced negative affect without positive-affect recovery.

General Discussion

Across two studies of a national midlife panel, this research asked whether adults retain AI-era core work skills as their personality profiles potentially shift over time. Four personality profiles were recovered and validated across independent maximum-likelihood implementations; Distressed membership declined markedly over two decades with little new onset; and among adults who left the Distressed profile basic psychological needs distinguished recovery-oriented transitions that preserved future-of-work skill criteria from disengagement-oriented transitions that did not. Current results extend person-centered research that treats latent profiles as configural structures to be discovered and replicated before outcome or transition hypotheses are tested (Gabriel et al., 2015; Zhang et al., 2024). Likewise, results contribute to latent profile and transition research examining how employees utilize psychological resources over time and why transition patterns carry distinct implications for adjustment (Chawla et al., 2020; Kanitz et al., 2024; Kleshinski et al., 2024; Slaughter et al., 2021; Vaziri et al., 2020). These studies link nationally representative midlife personality profiles to future-of-work skill criteria and model asymmetric profile transitions across twenty years. Finally, the current studies contribute to previous literature by demonstrating that the basic psychological needs predict recovery- versus disengagement-oriented transitions among Distressed movers.

Six conclusions can be drawn from the results of the current studies. First, latent personality profiles based on Big Five and Agency indicators were confirmed in a national midlife panel and converged across independent maximum-likelihood implementations (99.5% classification agreement). Independent cross-sectional re-enumeration reproduced the four-profile granularity and the dominant Resilient prototype, whereas the Distressed profile was recovered as a longitudinal state from the joint transition model rather than as a recurring cross-sectional cluster. Separation among profiles was predominantly elevational rather than configural

(Morin & Marsh, 2015), and profile membership did not improve prediction beyond continuous traits, consistent with prior latent-profile work (Costa et al., 2002; Meyer & Morin, 2016; Spurk et al., 2020). Both profiles and continuous traits were related to the criteria. However profile membership did not outperform the traits, so profiles are interpreted descriptively and developmentally rather than as a claim of predictive superiority.

Second, the enumeration correction reinforces long-standing guidance about local optima rather than introducing a new method. That finite-mixture and latent-transition likelihoods are non-concave and can trap estimation at a local solution is well established, as is the remedy of estimating each model from many random start values and confirming that the best log-likelihood replicates (Hipp & Bauer, 2006; McLachlan & Peel, 2000; Nylund-Gibson & Choi, 2018). The present case adds a cautionary demonstration: the initial three-class solution passed split-half reproduction, equal-covariance sensitivity, and external replication in the Refresher sample, yet was still a local optimum, surfaced only when the identical model was re-estimated with extensive random multistarts and cross-checked in a second, independently developed maximum-likelihood program. Replication of a single solution therefore validates that solution's neighborhood of the likelihood surface, not its global optimality. Two practices follow. Person-centered longitudinal studies should report multistart protocols (starts attempted and best-log-likelihood replication count) as routinely as fit indices (Hipp & Bauer, 2006; Nylund-Gibson & Choi, 2018), and, where feasible, treat re-estimation in an independently developed program and classification agreement as an inexpensive robustness and reproducibility check (Meyer & Morin, 2016; Muthén & Muthén, 2017).

Third, profile transitions were asymmetric over two decades. Distressed membership declined substantially while entry into distress was rare, consistent with episodic rather than

accumulating distress in this panel (Specht et al., 2014). Skill-criterion change was more strongly associated with within-person profile elevation than with discrete transition coding, whereas analytic performance showed only small change.

Fourth, the four-profile solution revealed a dissociation between psychosocial skill criteria (highest in the Resilient profile) and personal income, occupational prestige, and analytic performance (highest in the Antagonistic profile). Profiles were therefore not uniformly adaptive across criterion domains, paralleling the ambivalent and mixed configurations emphasized in recent person-centered work (Kanitz et al., 2024; Zhang et al., 2024). Antagonistic members reported life satisfaction comparable to Distressed members despite higher income and prestige, consistent with SDT goal-contents predictions (Kasser & Ryan, 1996; Niemiec et al., 2009).

Fifth, satisfaction of the three BPNs and supplementary purpose in life predicted transition direction among Distressed movers, with purpose reaching conventional significance and autonomy and relatedness showing marginal trends. Descriptive likelihood models showed no parallel associations. Each need also showed a distinct pattern of unique associations with the skill criteria (Deci & Ryan, 2000; Ryan & Deci, 2017).

Sixth, the asymmetry of distress decline, the recovery-versus-disengagement divergence, and the need-specific pattern of SDT need-criterion associations imply that sustaining the psychosocial skill criteria named in employer core-skills surveys may depend less on aggregate strain prevalence than on the quality of developmental transitions out of distress. Research and practice implications are developed below.

Implications

Theoretical and Research Implications

The central question motivating this work is whether workers under transition-induced strain emerge with the core and rising workforce skills prioritized for the AI transition. Study 1 fixed that skill-criterion set from the future-of-work skill framework (World Economic Forum, 2025). Study 2 showed that the answer is dependent on pathway. Transitions out of the Distressed profile primarily involved either recovery that preserved generative and prosocial skill expression or disengagement that did not. For research, this argues for treating transition quality, not distress prevalence alone, as a primary longitudinal outcome in future-of-work studies (Bergman & Magnusson, 1997; Chawla et al., 2020). Person-centered latent transition models are well suited to that question because they preserve configural meaning while tracking asymmetric flows across multiyear intervals relevant to structural labor-market change (Collins & Lanza, 2010; Howard et al., 2020; Huang et al., 2024).

A second research implication concerns the divergence between psychosocial skills (highest in the Resilient profile) and income, prestige, and analytic performance (highest in the Antagonistic profile). Future research should therefore model skill maintenance under AI transition as multi-domain rather than assuming that productivity, prestige, or technical performance are proxies that subsume the skills employers rank as rising priorities (World Economic Forum, 2025; Kasser & Ryan, 1996).

Third, the basic psychological needs predicted transition direction among Distressed movers but not transition likelihood, and relationships between needs and skill criteria were need-specific across the full criterion space. That pattern challenges undifferentiated resource and buffering accounts and supports intervention theories that match specific needs to specific criteria (Ryan & Deci, 2017). Researchers should test matched need-criterion hypotheses in organizational field studies and experiments. Finally, purpose here was general purpose in life.

Workplace-specific meaning and job-based purpose remain priority constructs for the AI transition literature (Park, 2010; Steger et al., 2012).

Practical Implications for Organizations and Leaders

If AI-era workforce planning aims to maintain or increase human workforce skills rather than only automate tasks, leaders should focus on the quality of transitions out of strain at least as much as on reducing strain itself. These findings align with work-design scholarship arguing that digital transformation must be approached sociotechnically, jointly optimizing technology and the design of human work rather than treating upskilling as a substitute for job redesign (Gagné et al., 2022; Parker & Grote, 2022).

Three levers map directly onto recovery-oriented versus disengagement-oriented transitions. First, decision authority and autonomy-supportive management mattered for transition direction but not for whether Distressed adults left at all. Leaders can act on that distinction immediately by preserving meaningful choice, rationale transparency, and human override in AI-augmented workflows rather than assuming that distress will self-correct once tools are deployed (Deci et al., 2017; Gallup, 2024). Second, purpose in life predicted recovery-oriented direction among movers, with autonomy and relatedness showing parallel trends. Near-term managerial practice should therefore connect role redesign to contribution and developmental meaning rather than only efficiency goals, for example, by clarifying how redesigned work creates value for others, supports mentoring or generative roles, and fits a worker's sense of direction (McAdams & de St. Aubin, 1992; Steger et al., 2012). Third, relatedness showed domain-specific ties to generativity and reappraisal, with competence contributing to early-interval self-regulatory and generative gains. Skill-building programs tied to AI adoption should emphasize mastery experiences that expand human judgment, coaching,

and cross-role collaboration (the future-of-work rising skills of creative thinking, leadership, and social influence) rather than narrow tool training alone (World Economic Forum, 2025).

Leaders should also monitor the recovery-versus-disengagement distinction explicitly. Disengaged movers in the diary data reduced negative affect without restoring positive affect. At the skill-criterion level, disengagement preserved distress reduction while sacrificing generativity. That pattern cautions against interpreting lower burnout scores, withdrawal from conflict, or muted complaint as successful adaptation during restructuring. Prospective diary evidence further suggests that chronic distress was forecast by repeated interpersonal conflict exposure rather than by appraised severity alone, implying that managers should reduce predictable daily friction (unclear accountability, unmanaged team conflict, and work-family spillover) rather than only train resilience (Almeida, 2005).

These priorities do not guarantee causal skill-criterion gains from specific organizational interventions, because the non-experimental panel design does not manipulate job design or need-support treatments. The priorities nonetheless translate the study's central finding into implementable terms. Under technological transition, workforce skills are more likely to be preserved when strained workers recover agency and generative investment, not when they merely disengage.

Limitations

This study is a secondary analysis of a nationally representative longitudinal panel (Brim et al., 2004; Radler & Ryff, 2010) with several limitations. First, the design is correlational. The basic psychological needs, decision authority, and purpose were not experimentally manipulated, so the profile-transition and need-direction findings establish temporal ordering and baseline control but not causal identification of organizational levers and therefore confounds cannot be

ruled out (Angrist & Pischke, 2009; West & Thoemmes, 2010). Experimental or quasi-experimental designs would be needed to test whether providing organizational resources changes transition direction. Second, the transition-role, need-criterion, and diary analyses are exploratory so while their consistency across the two decade-long intervals is encouraging it is not confirmatory. Subsequent preregistered replication research is warranted. Third, several short-form indicators have low internal consistency in the total sample, notably the three-item Ryff purpose ($\alpha = .29$) and autonomy ($\alpha = .45$) scales, which attenuates the need-direction associations central to the self-determination argument. These need indicators are archival well-being measures (Ryff subscales and personal mastery) rather than a dedicated basic-need-satisfaction instrument (e.g., the Basic Psychological Need Satisfaction and Frustration Scale), so construct coverage of the three needs is approximate. A supplementary analysis using the MIDUS seven-item Ryff forms available at MIDUS 2 showed larger associations in the same direction (see Tables S2 and S3 in the online supplemental materials), consistent with attenuation rather than a scoring artifact. Full-length need measures with higher internal consistency would improve the power of statistical tests (McNeish, 2018). Fourth, class enumeration favored progressively more classes on BIC through $k = 5$. The four-class solution was selected on stability and interpretability and reproduced across two engines, but only a small share of random starts recovered its best-fitting values, and classification entropy was moderate (.71); additional indicators and confirmatory replication would more firmly establish the profile structure. Finally, the economic figures use the human-capital method (Rice, 1967; Stewart et al., 2003) and partly reflect wage composition, so they are illustrative, panel-based projections rather than treatment-effect estimates (Cook & Campbell, 1979).

Conclusion

This research asked whether the personality resources that underlie core workforce skills (World Economic Forum, 2025) are sustained when adults face strain across the two decades preceding the current AI-era transition. In a national panel, midlife personality profiles were largely elevational rather than configural and did not add incremental prediction beyond continuous traits, so their contribution is descriptive and developmental rather than a claim of predictive superiority. That contribution is nonetheless substantive. Distressed configurations became markedly less common over twenty years, and the manner in which adults left distress determined the outcome. Recovering agency and generative investment, rather than merely disengaging, distinguished preservation of psychosocial skills from their erosion, with the basic psychological needs and purpose predicting the direction of that transition. For organizations navigating AI-era transition, the results reframe the objective from lowering the prevalence of strain to improving the quality of recovery from it.

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Table 1

Descriptive Statistics and Correlations Among Study 1 MIDI Personality Composites and Study 2 Distal Outcomes (MIDUS 2, N = 3,970)

Variable	M	SD	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1. Neuroticism	2.07	0.62	(.74)													
2. Extraversion	3.10	0.57	-.20	(.76)												
3. Openness	2.90	0.54	-.21	.51	(.77)											
4. Agreeableness	3.45	0.50	-.11	.50	.33	(.80)										
5. Conscientiousness	3.46	0.45	-.19	.25	.28	.27	(.58)									
6. Agency	2.61	0.66	-.10	.51	.52	.07	.22	(.81)								
7. Sense of control	5.53	1.00	-.41	.36	.37	.16	.35	.34	(.87)							
8. Reappraisal	3.05	0.61	-.29	.43	.42	.35	.25	.29	.41	(.78)						
9. Generativity	16.99	3.85	-.15	.41	.46	.33	.23	.35	.30	.39	(.85)					
10. Mastery	5.74	1.02	-.25	.34	.35	.16	.26	.35	.74	.40	.26	(.73)				
11. Autonomy	16.54	3.08	-.25	.24	.32	.13	.23	.32	.36	.29	.22	.28	(.45)			
12. Relatedness	16.85	3.80	-.33	.44	.26	.41	.27	.19	.47	.35	.33	.30	.29	(.63)		
13. Job satisfaction	7.47	2.15	-.23	.19	.12	.12	.19	.12	.34	.17	.15	.21	.10	.26	—	
14. Life satisfaction	7.77	1.24	-.34	.33	.21	.21	.30	.17	.48	.29	.22	.34	.19	.42	.76	–

Note. Diagonal entries are Cronbach's alpha. Variables 1-6 are unit-weighted MIDI scale scores on the original 4-point adjective metric (1 = not at all, 4 = a lot); LPA was estimated on within-wave z-scored ESEM factor scores, for which Pearson correlations are equivalent. Possible score ranges: sense of control and mastery, 1–7; reappraisal, mean of four 1–4 items; generativity, 6–24; autonomy and relatedness (Ryff three-item sums), 3–21; job and life satisfaction, 0–10. Higher scores indicate more of the construct. Sample is the Study 1 primary analytic sample (complete MIDUS 2 personality indicators); distal-outcome ns = 3,685-3,970 owing to item nonresponse. Values on the diagonal are Cronbach's α from MIDUS psychometric reports (Lachman & Weaver, 1997, 1998; Mroczek & Kolarz, 1998; Ryff & Keyes, 1995). Job and life satisfaction are single items (em dash on the diagonal); single-item job satisfaction has estimated minimum reliability of .67-.70 (Wanous et al., 1997). Positive affect, negative affect, purpose in life, and economic criteria are omitted here to conserve space but were included in profile-outcome models (Tables 4 and 5).

Table 2

Fit Statistics for the Joint Latent Markov Model (k = 2–5)

k	LL	FP	BIC	Entropy
2	-86,671.6	38	173,676	.80
3	-85,524.3	53	171,513	.83
4	-84,708.5	72	170,048	.76
5	-84,011.3	95	168,856	.80

Note. LL = log-likelihood; FP = free parameters; BIC = Bayesian information criterion. Each model was estimated by maximum likelihood with 31 random start sets (LMest; Bartolucci et al., 2017). Entropy is relative entropy computed from the best-LL solution's person–wave state posteriors (forward–backward). The retained k = 4 solution was confirmed by an independent maximum-likelihood implementation (Mplus 8.8; LL = −84,834.97, BIC = 170,301, entropy = .71; modal classification agreement between implementations = 99.5%). The full k = 2–5 series was re-estimated with that implementation under the analogous common within-class covariance specification and reproduced the fit ordering, with monotonically declining BIC (173,950, 171,596, 170,301, and 169,062 for k = 2–5), correspondingly declining log-likelihood (−86,808.2, −85,565.8, −84,835.0, and −84,114.5), and comparable classification quality (relative entropy = .71, .75, .71, and .75); the four-profile decision was unchanged across implementations. An extended 1,500-start maximum-likelihood run for k = 4 recovered the best LL from two independent seeds. The earlier k = 3 solution was a shallower local optimum relative to the multistart best LL shown here. At k = 5, additional starts altered class structure, so that solution was not retained.

Table 3

Latent Class Profile Means on Anchored Personality Factor Scores (Joint Latent Markov Model, k = 4)

Construct	Resilient	Distressed	Reserved	Ant.
Neuroticism	-0.53	0.71	-0.24	-0.18
Extraversion	0.12	0.20	-0.51	-0.93
Openness	-0.04	0.18	-0.42	-0.58
Agreeableness	0.49	0.54	-0.84	-2.16
Conscientiousness	0.28	0.11	-0.14	-0.35
Agency	-0.10	0.10	-0.19	-0.19
M1 (%)	36.4	29.5	29.0	5.1
M1 n	2,343	1,899	1,867	328
M2 (%)	54.3	8.8	31.0	5.9
M2 n	3,496	567	1,996	380
M3 (%)	53.3	7.2	34.7	4.8
M3 n	3,431	464	2,234	309

Note. Entries are latent class profile means in longitudinally anchored factor-score SD units (M1 metric), from the retained maximum-likelihood solution. A second, independently developed maximum-likelihood implementation reproduced these means to within rounding (maximum absolute difference approximately 0.03). Wave-specific prevalence percentages and sample sizes (prevalence × N) are model-estimated marginal quantities from the confirmed joint latent Markov model (FIML; N = 6,438). Ant. = Antagonistic profile.

Table 4

Future-of-Work Core Skills by Profile (k = 4): Adjusted Means at MIDUS 2

Skill criterion	Resilient	Distressed	Reserved	Ant.	p
Sense of control	5.70_{a}	4.99_{c}	5.35_{b}	5.07_{b}	< .001
Reappraisal	3.18_{a}	2.98_{b}	2.86_{c}	2.67_{d}	< .001
Generativity	17.76_{a}	17.46_{a}	15.67_{b}	14.47_{c}	< .001
Purpose in life	16.51_{a}	15.55_{b}	15.83_{b}	14.81_{b}	< .001
Mastery	5.87_{a}	5.51_{b}	5.56_{b}	5.29_{b}	< .001
Personal income	9.48_{b}	9.44_{b}	9.56_{a}	9.76_{a}	< .001
Occupational prestige	48.15_{a}	47.83_{b}	48.42_{a}	49.04_{a}	.006
BTACT composite	-0.02_{b}	-0.05_{b}	0.01_{a}	0.03_{a}	< .001

Note. Possible score ranges: sense of control and mastery, 1–7 (higher = greater control/mastery); reappraisal, mean of four 1–4 items; generativity, Loyola six-item sum 6–24; purpose in life, Ryff three-item sum 3–21. Personal income is log-transformed dollars. BTACT composite scores are within-sample standardized (z). Occupational prestige is Duncan SEI. Means adjusted for age, sex, and education (Ns = 1,604-2,522); p values are omnibus profile-effect F tests from covariate-adjusted linear models under modal profile assignment. Values in the p column report exact p when p >= .001 and as < .001 otherwise. Subscript letters denote pairwise differences at p < .05 from Tukey-adjusted comparisons following significant omnibus profile effects; profiles sharing a subscript do not differ on that outcome. Psychosocial skill criteria were highest in the Resilient profile; personal income, occupational prestige, and the BTACT composite were highest in the Antagonistic profile (Reserved ranked second on income and prestige) -- the skill-criterion dissociation (Figure 1, Panel C). Ant. = Antagonistic profile.

Table 5

Well-Being Outcomes by Profile (k = 4): Adjusted Means at MIDUS 2

Outcome	Resilient	Distressed	Reserved	Ant.	d	p
Job satisfaction	7.64_{a}	6.83_{c}	7.26_{b}	7.22_{b}	-.20	< .001
Life satisfaction	7.96_{a}	7.36_{b}	7.53_{b}	7.35_{b}	-.51	< .001
Positive affect	3.56_{a}	3.03_{c}	3.27_{b}	3.12_{c}	-.65	< .001
Negative affect	1.43_{c}	2.11_{a}	1.57_{b}	1.58_{b}	.27	< .001
Personal income	9.48_{b}	9.44_{b}	9.56_{a}	9.76_{a}	.10	< .001

Note. Possible score ranges: job and life satisfaction, 0–10; positive and negative affect, 1–5. Personal income is log-transformed dollars. Higher scores indicate more of the construct unless noted. Means adjusted for age, sex, and education (Ns = 2,324-2,525); p values are omnibus profile-effect F tests from covariate-adjusted linear models under modal profile assignment; d is the Antagonistic-Resilient contrast in residual-SD units. Values in the p column report exact p when p >= .001 and as < .001 otherwise. Subscript letters denote pairwise differences at p < .05 from Tukey-adjusted comparisons following significant omnibus profile effects. The Antagonistic profile pairs the highest income with life satisfaction indistinguishable from the Distressed profile (7.35 vs. 7.36). A profile-by-income moderation analysis showed income unrelated to life satisfaction within the Antagonistic profile (b = -0.02, p > .05) but positively related within the Distressed profile (b = +0.35, p < .001), relative to the Resilient slope. Ant. = Antagonistic profile.

Table 6

SDT Basic Psychological Needs and Transitions Out of the Distressed Profile

Predictor	Leaving full sample	Leaving employed	Direction full sample	Direction employed
Autonomy	1.07	0.99	1.20†	1.16
Competence	1.09	1.09	1.01	1.01
Relatedness	1.09	1.17	1.15	1.22†
Purpose in life	0.88	0.91	1.23*	1.25*
Decision authority	—	1.04	—	1.17

Note. Entries are adjusted odds ratios per 1 SD increase in the predictor from the k = 4 core model for the MIDUS 1 to MIDUS 2 interval. Leaving = odds of leaving versus staying in the Distressed profile (full n = 1,063; employed-at-M1 n = 780). Direction = odds of recovery versus disengagement among movers to Resilient or Reserved (full n = 898; employed n = 657). Autonomy and relatedness are Ryff Psychological Well-Being subscales; competence is personal mastery; decision authority is the Karasek job-characteristic scale (employed subsample only). Models adjust for age, sex, and education. Decision authority estimated among employed respondents only (likelihood n = 731; direction n = 618). Em dash indicates estimate not applicable for that subsample. * $p < .05$. † $p < .10$.

Table 7

Model-Estimated Latent Profile Transition Probabilities (k = 4 Joint Latent Markov Model)

Interval	Origin profile	Resilient	Distressed	Reserved	Ant.
M1→M2	Resilient	.852	.000	.139	.008
	Distressed	.520	.279	.184	.017
	Reserved	.264	.020	.640	.076
	Antagonistic	.048	.000	.384	.568
M2→M3	Resilient	.818	.003	.174	.006
	Distressed	.079	.750	.172	.000
	Reserved	.259	.014	.678	.050
	Antagonistic	.023	.010	.465	.502

Note. Cell entries are row-to-column transition probabilities from the confirmed joint latent Markov model (FIML). Rows sum to 1.00 within each origin profile and interval. Transitions out of the Distressed profile at M1→M2 flow predominantly to Resilient (.520) with a secondary flow to Reserved (.184). These probabilities are visualized as the ribbon widths in Figure 1, Panel B. Ant. = Antagonistic profile.

Table 8

Two Transition Pathways From Distress: M2 Skill-Criterion Levels of M1-Distressed Movers Versus Stayers

Skill criterion	Recovered vs. stayed		Disengaged vs. stayed	
	β	*p*	β	*p*
Sense of control	+0.31	< .001	-0.04	.60
Mastery	+0.16	.009	-0.17	.046
Reappraisal	+0.21	< .001	-0.21	.008
Generativity	+0.04	.48	-0.42	< .001

Note. Entries are baseline-adjusted standardized coefficients (β) in SD units, comparing M2 skill-criterion levels of M1-Distressed movers with stayers. Possible score ranges: sense of control and mastery, 1–7 (higher = greater control/mastery); reappraisal, mean of four 1–4 items; generativity, Loyola six-item sum 6–24. N = 1,100-1,119 M1-Distressed adults. Recovery raises control-family skill criteria with generativity preserved; detachment leaves control unchanged and costs generativity and reappraisal.

Table 9

Mean Change Scores (M1→M2) on Skill Criteria by Distressed-Profile Transition Pathway

Pathway	Outcome	n	ΔM	SE	p
Recovered	Sense of control	347	+0.15	0.06	.007
	Mastery	347	-0.06	0.06	.327
	Reappraisal	347	0.00	0.03	.982
	Generativity	343	-0.02	0.16	.887
	Job satisfaction	323	0.01	0.16	.954
Disengaged	Sense of control	120	-0.11	0.09	.227
	Mastery	120	-0.24	0.10	.018
	Reappraisal	120	-0.23	0.06	< .001
	Generativity	118	-0.58	0.33	.076
	Job satisfaction	113	0.00	0.24	1.00
Stayed distressed	Sense of control	145	-0.13	0.08	.101
	Mastery	145	-0.20	0.10	.041
	Reappraisal	144	0.00	0.05	.952
	Generativity	138	0.12	0.30	.686
	Job satisfaction	130	-0.22	0.28	.430

Note. Possible score ranges: sense of control and mastery, 1–7; reappraisal, mean of four 1–4 items; generativity, 6–24; job satisfaction, 0–10. ΔM is change on the same metric at MIDUS 2 minus MIDUS 1. Recovered = Distressed at MIDUS 1 to Resilient at MIDUS 2; Disengaged = Distressed to Reserved; Stayed distressed = Distressed at both waves. ΔM = paired mean change from MIDUS 1 to MIDUS 2; SE = standard error of the mean change. Recovery raises sense of control relative to disengagement and chronic distress; disengagement costs reappraisal and mastery (cf. Table 8).

Table 10

Unique Standardized Effects of SDT Basic Needs on Residualized Skill-Criterion Change (Full Sample)

Outcome	M1→M2 Aut.	M1→M2 Comp.	M1→M2 Rel.	M1→M2 Purp.	M2→M3 Aut.	M2→M3 Comp.	M2→M3 Rel.	M2→M3 Purp.
Sense of control	.04*	.08*	.03	.07*	.04*	.01	.06*	.05*
Reappraisal	.04*	.05*	.05*	.02	.05*	.05*	.06*	.01
Generativity	.03	.03*	.06*	.02	.02	.03	.07*	.04*
Personal income	-.01	.01	-.01	.04*	-.04	.03	.05*	.07*
Occupational prestige	.00	.01	.00	.05*	.05*	.01	.01	-.01
BTACT analytic	—	—	—	—	.01	-.02	.01	.03*

Note. Possible score ranges: sense of control and mastery, 1–7 (higher = greater control/mastery); reappraisal, mean of four 1–4 items; generativity, Loyola six-item sum 6–24; purpose in life, Ryff three-item sum 3–21. Personal income is log-transformed dollars. BTACT analytic scores are within-sample standardized (z). Occupational prestige is Duncan SEI. BTACT analytic composite is a within-sample standardized score. Column headers are wave interval (M1→M2 or M2→M3) over Aut. = autonomy, Comp. = competence, Rel. = relatedness, and Purp. = purpose. Autonomy, competence, relatedness, and purpose entered jointly with baseline outcome, age, sex, and education in each model; entries are unique standardized β coefficients on wave-standardized outcomes. BTACT analytic composite is available for MIDUS 2 to MIDUS 3 only. Em dash indicates outcome not estimated in that interval. * $p < .05$. † $p < .10$.

Table 11

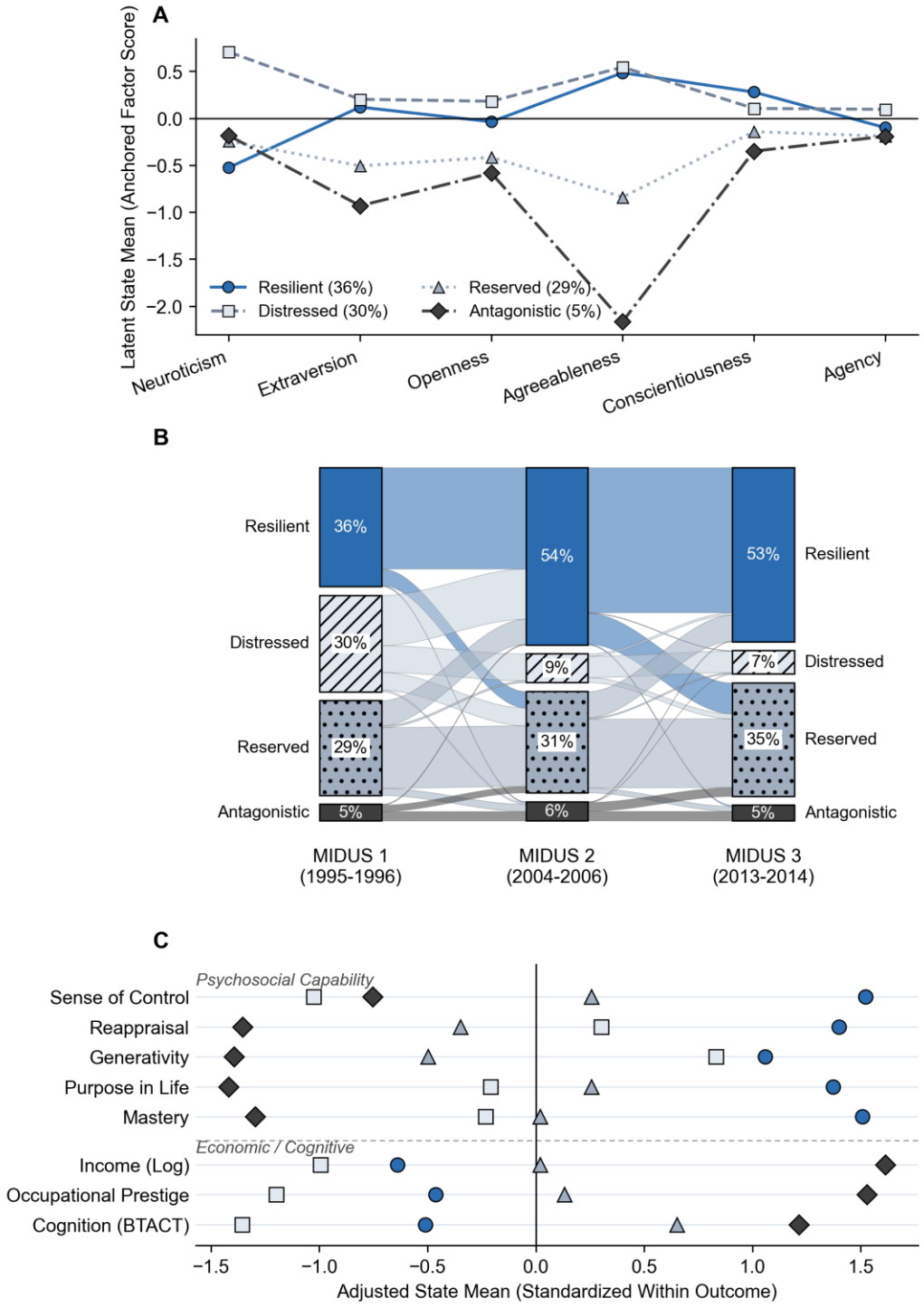
Daily Stress Ecology by k = 4 Profile (Three Diary Samples)

Sample	Measure	n	Resilient	Distressed	Reserved	Ant.	p
NSDE II	Stressor days (proportion)	1,106	.375	.444	.381	.481	.009
	Interpersonal conflict days (proportion)	1,106	.211	.283	.208	.298	.002
	Stressor severity	992	1.726	1.997	1.705	1.770	.026
	Evaluative threat	960	.281	.392	.355	.500	.0004
	Daily negative affect	1,106	.158	.312	.185	.218	< .001
	Daily positive affect	1,106	2.898	2.543	2.557	2.372	< .001
	Affective reactivity	966	.152	.202	.177	.161	.290
M3 diary	Stressor days (proportion)	463	.369	.462	.416	.412	.220
	Interpersonal conflict days (proportion)	463	.199	.281	.218	.241	.290
	Stressor severity	400	1.782	2.120	1.753	1.475	.005
	Evaluative threat	400	.269	.278	.416	.523	.002
	Daily negative affect	463	.138	.292	.195	.190	.0001
	Daily positive affect	463	2.835	2.398	2.472	2.286	< .001
	Affective reactivity	389	.121	.222	.103	.154	.075
Refresher	Stressor days (proportion)	745	.395	.492	.417	.425	.010
	Interpersonal conflict days (proportion)	745	.229	.282	.243	.270	.093
	Stressor severity	691	1.655	1.975	1.716	1.684	< .001
	Evaluative threat	691	.351	.514	.429	.506	.004
	Daily negative affect	745	.146	.385	.227	.251	< .001
	Daily positive affect	745	2.878	2.324	2.356	2.160	< .001
	Affective reactivity	668	.139	.250	.153	.167	.0002

Note. Stressor severity, 0–3. Daily positive and negative affect are person-level means on 1–4 scales. Stressor days and interpersonal conflict days are proportions of valid diary days. Evaluative threat is the person-level mean of combined self- and other-evaluative threat items (0–3). Person-level diary aggregates require at least three valid days. NSDE II and M3 diary samples use modal k = 4 assignments (M2 and M3) from the k = 4 core model; Refresher uses nearest-centroid classification on anchored six-factor scores. Omnibus p values are one-way ANOVA tests across observed profiles in each sample. Values in the p column report exact p when p >= .001 and as < .001 otherwise. Ant. = Antagonistic profile.

Figure 1

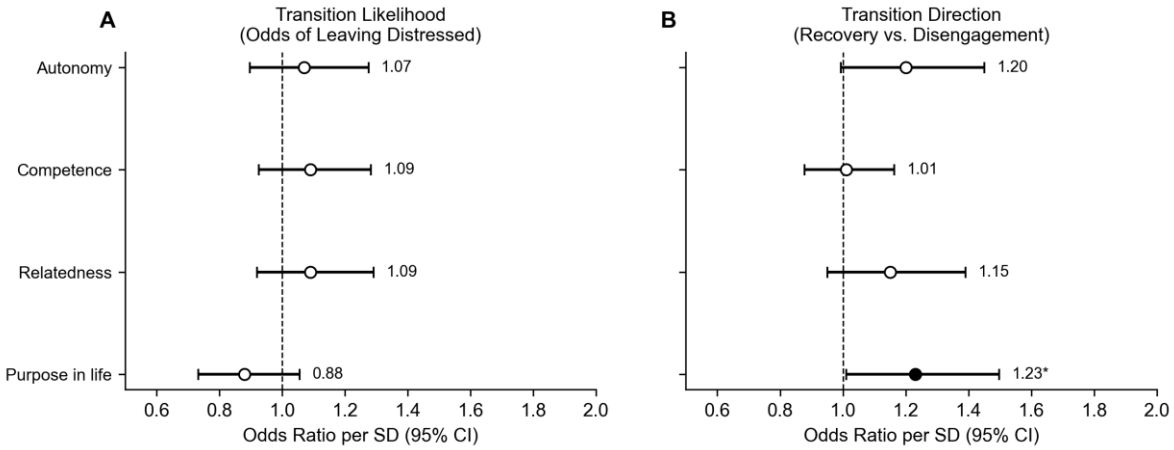
The Four-Profile Personality Solution: Latent Class Means, Transitions, and Skill-Criterion Dissociation



Note. Panel A: latent class profile means on longitudinally anchored factor scores (M1 metric), with baseline prevalences in the legend. Panel B: joint latent Markov model profile prevalences across the three MIDUS waves; node heights and percentages are model-estimated wave-specific prevalences (Distressed 29.5% to 8.8% to 7.2%), and ribbon widths are the model-estimated transition probabilities (Table 7) scaled by each origin profile's prevalence. Panel C: future-of-work skill patterns adjusted for age, sex, and education and standardized within outcome; distances are comparable across rows but not across psychosocial skill criteria and income, prestige, and analytic outcomes in raw units. Markers and shading identify the same profiles in all panels. BTACT = Brief Test of Adult Cognition by Telephone.

Figure 2

SDT Basic Psychological Needs and Transitions Out of the Distressed Profile (k = 4 Core Model, MIDUS 1 to MIDUS 2)



Note. Panel A: odds of leaving the Distressed profile; Panel B: odds of recovery versus disengagement among movers. Estimates are for the primary M1-to-M2 distress-transition interval (n = 898 movers to Resilient or Reserved). A parallel M2-to-M3 battery was not plotted because only 53 Distressed-profile movers remained at MIDUS 2, yielding unstable direction estimates; exploratory M2-to-M3 coefficients appear in the supplement. Autonomy and relatedness are Ryff Psychological Well-Being subscales (Ryff, 1989; Ryff & Keyes, 1995); competence is personal mastery (Lachman & Weaver, 1998); purpose in life is the Ryff purpose subscale (Ryff, 1989). OR = odds ratio per SD, adjusted for age, sex, and education. Error bars are approximate 95% confidence intervals reconstructed from reported p values. Filled markers denote significance at $p < .05$; open markers are not. * $p < .05$. † $p < .10$.

Appendix A

A Priori Mapping of Future of Jobs Core Skills to MIDUS Measures

The outcome space was fixed by an external employer framework before any distal-outcome model was run: each skill named in the World Economic Forum's Future of Jobs core-skills list (World Economic Forum, 2025) was mapped to the closest validated MIDUS instrument, and every mapped criterion was retained and reported regardless of result (including null and inconvenient findings, e.g., the near-zero BTACT effects). With the BTACT cognitive battery included, the criterion set covers all five top Future of Jobs core skills and the principal rising skills named for 2025-2030.

Table A1

Future of Jobs Skill-to-MIDUS Criterion Mapping

WEF skill	MIDUS operationalization	Source
Analytical thinking (WEF #1)	BTACT composite, executive functioning, episodic memory	Tun & Lachman (2006)
Resilience, flexibility, agility (WEF #2, rising)	Sense of control; personal mastery; perceived constraints (reversed); positive reappraisal	Lachman & Weaver (1998)
Leadership and social influence (WEF #3, rising)	Agency (MIDI indicator); decision authority; supervisory status and span	Lachman & Weaver (1997); Karasek (1979)
Creative thinking (WEF #4, rising)	Generativity; personal growth; Openness (indicator)	McAdams & de St. Aubin (1992); Ryff (1989)
Motivation and self-awareness (WEF #5)	Self-acceptance; personal mastery; personal growth	Ryff (1989); Lachman & Weaver (1998)
Curiosity and lifelong learning (WEF, rising)	Openness (indicator); personal growth	Lachman & Weaver (1997); Ryff (1989)
Economic standing (labor-market criterion)	Personal income; occupational prestige (SEI)	MIDUS employment module

Note. Rankings refer to the Future of Jobs Report 2025 skills lists (World Economic Forum, 2025). The mapped criteria span all five of the top core skills (those judged essential in the current workforce); three of these also appear on the rising-skills list (skills growing most in importance through 2030) and are labeled rising, and curiosity and lifelong learning is included as a further rising skill. The criterion set is therefore predominantly core, with substantial overlap onto the rising list. Constructs labeled (indicator) are variables used to estimate the latent profiles

(e.g., Openness, Agency), not distal outcomes predicted from profile membership; they appear only to document construct coverage and were not used as outcomes, which avoids circularity. Economic standing is a supplementary labor-market criterion (income and occupational prestige), not a Future of Jobs skill. The full mapping was fixed a priori, before any distal-outcome model was estimated.

Appendix B

Data Transparency Statement

This manuscript reports a secondary analysis of publicly available data from the Midlife in the United States (MIDUS) study. Data were obtained through the MIDUS data portal (Colectica; https://midus.colectica.org) maintained by the MIDUS project, with study-level cross-reference to the Inter-university Consortium for Political and Social Research (ICPSR) deposits listed below. No new human-participants data were collected for this manuscript. MIDUS data are subject to the MIDUS conditions of use and ICPSR terms of use and, where applicable, restricted-use agreements.

The core survey deposits and their ICPSR digital object identifiers are MIDUS 1 (ICPSR 2760; https://doi.org/10.3886/ICPSR02760), MIDUS 2 (ICPSR 4652; https://doi.org/10.3886/ICPSR04652), MIDUS 3 (ICPSR 36346; https://doi.org/10.3886/ICPSR36346), and MIDUS Refresher 1 (ICPSR 36532; https://doi.org/10.3886/ICPSR36532). The daily-diary (National Study of Daily Experiences) subsamples analyzed in Study 2 are the MIDUS 2 Daily Stress Project (ICPSR 26841; https://doi.org/10.3886/ICPSR26841), the MIDUS Refresher Daily Diary (ICPSR 37083; https://doi.org/10.3886/ICPSR37083), and the MIDUS 3 Daily Diary (ICPSR 38529; https://doi.org/10.3886/ICPSR38529). Data were accessed through the MIDUS Colectica portal; because ICPSR deposits are versioned, the most recent version available at the time of access was used (following Kirkman & Chen, 2011, and Journal of Applied Psychology data-transparency requirements).

MIDUS data were used only for the analyses described in this manuscript, no attempt was made to identify individual respondents, and restricted files were not redistributed. Analysis

code and derived output tables supporting this draft are maintained in a project repository. Requests for code or analysis logs should be directed to the corresponding author.

Supplemental Material

The following tables are intended for the online supplemental materials.

Table S1

Cross-Sectional LPA Enumeration at MIDUS 2 (ESEM Factor Scores, Employed Subsample, N = 2,507)

No. of profiles	LL	BIC	Entropy	Smallest class, %
1	-9,296.0	18,783.1	—	100.0
2	-8,967.9	18,325.2	.74	36.5
3	-8,872.4	18,332.2	.78	18.3
4	-8,809.4	18,404.5	.70	13.0
5	-8,785.4	18,554.5	.66	15.5

Note. LL = log-likelihood; BIC = Bayesian information criterion. Models used an unrestricted (class-specific) within-class covariance structure on within-wave z-scored ESEM factor scores (half-sample enumeration grid). This cross-sectional enumeration on the employed MIDUS 2 subsample is distinct from the joint k = 4 latent Markov model (Table 2); its class solutions and smallest-class percentages (e.g., 13.0% at G = 4) do not correspond to the joint model's profiles or prevalences. Selection rule: lowest BIC subject to entropy ≥ .70 and smallest class ≥ 5%, which retained G = 2. Split-half Tucker congruence for the G = 2 means was .99/.98; profile membership added no meaningful out-of-fold incremental validity over continuous traits (see Incremental Validity of Profile Membership).

Table S2

Sensitivity: Resilient Versus Distressed Separations on Three-Item Versus Seven-Item Ryff Scales at MIDUS 2

Measure	d (3-item)	d (7-item)
Autonomy	0.40	0.59
Purpose in life	0.26	0.57
Relatedness (positive relations)	0.59	0.67

Note. Cohen's d for Resilient minus Distressed on MIDUS 2 modal k = 4 membership (Resilient n ≈ 1,409; Distressed n = 165). Three-item composites are the primary longitudinal forms (B1SPWBA1/B1SPWBU1/B1SPWBR1); seven-item forms (B1SPWBA2/B1SPWBU2/B1SPWBR2) are available at MIDUS 2 and later only. All contrasts $p < .01$; seven-item ds are larger in the same direction, consistent with attenuation from short-form unreliability.

Table S3

Sensitivity: Residualized MIDUS 2→MIDUS 3 Capability Change Predicted by Three-Item Versus Seven-Item Needs (Single-Need Models)

Need	Outcome	β (3-item)	β (7-item)
Autonomy	Sense of control	.055	.097
	Mastery	.048	.085
	Reappraisal	.069	.080
	Generativity	.047	.063
Purpose in life	Sense of control	.068	.130
	Mastery	.038	.100
	Reappraisal	.044	.083
	Generativity	.064	.105
Relatedness	Sense of control	.080	.082
	Mastery	.072	.080
	Reappraisal	.083	.089
	Generativity	.091	.103

Note. Entries are standardized betas from models of the form z(y_M3) ~ z(y_M2) + z(need) + age + sex + education (n ≈ 2,660-2,671). Competence (personal mastery) is unchanged across columns and is omitted. Seven-item autonomy/purpose/relatedness replace the three-item forms; signs are unchanged and absolute associations are generally larger. The primary H4 transition-direction models cannot be re-estimated with seven-item predictors because those forms are unavailable at MIDUS 1.